

EEL 3111C Lab Manual - Fall 2026

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Chapter 1

Lab 1 - Rules and Policies

The purpose of this laboratory is to provide a hands on experience for students in conjunction with the recorded lectures and problem sessions. While solving nodal equations and drawing circuits has a lot of utility, there is a great amount of understanding to be gained from building circuits, seeing them work, and then understanding why they work. The objectives of the laboratories are as follows:

- To verify concepts presented in the course.
- To enable students to gain skills in using basic electrical laboratory equipment such as meters, power supplies, and oscilloscopes, which will be need in subsequent electrical engineering laboratories and in industry.
- To make the students aware of the limits of the operation of components and equipment.
- To have the students gain skills in recording data and in reporting experimental results to effectively communicate what they have done and observed.
- To have the students become safety conscious in a laboratory environment.

1.1 Requirements

Each student is required to have the following:

- A laptop with Waveforms installed
- Digilent Analog Discovery Board
- Breadboard
- Wiring kit
- Lab parts kit

A Digilent Analog Discovery 3 for this course can be purchased from the UF bookstore. For those who don't want to purchase their own, the ECE department can lend students a Digilent Analog Discovery 3.

Any equipment lent to students by the department must be returned before the student's graduation date for EE major students. Non-EE major students must return the DAD by the end of the semester (the last day of lab). A breadboard, wiring kit, and lab parts kit will be provided during the first lab session. These lab materials do not need to be returned.

The lab manual is subject to change as errors are corrected, students are responsible for the newest version of the lab manual.

1.2 Structure

The laboratory portion of the course is made up of nine labs and a final project and takes place in MAEB 227. Every laboratory will be composed of an objective, list of materials, introduction, big picture, pre-lab requirements, in-lab requirements, and a write up.

1. **Objective:** States the goals for a given lab.
2. **Materials:** List the materials used for a lab.
3. **Introduction:** Introduces the overall theme of the lab and the supporting background information.
4. **Big Picture:** A block diagram of the final project showing how the circuits in this lab fit together in the final project.
5. **Pre-Lab Requirements:** Most of the lab is accomplished by the student at home. The requirements for the pre-lab will list out what needs to be completed by students prior to their entry into the lab.
 - Pre-lab quizzes are available on Canvas for 24 hours before the start of lab and must be completed by 8:30am on the day of your lab.
 - Pre-lab submissions to Canvas: tables, circuit diagrams and their resulting outputs, etc., must be compiled into a document and submitted to Canvas by 8:30am on the day of your lab. No late submissions are accepted, unless given permission by your lab TA.
 - Pre-lab circuit demonstrations must be ready within 15 minutes of the start of lab.
 - **Failure to complete the pre-lab on time, either the quiz, submission, or demonstration, will result in point deductions up to the discretion of the course instructor or lab TA for a given lab. Exceptions may be made by the course instructor or lab TA.**
6. **In-Lab Requirements:** Additional lab tasks that build on the work done in the pre-lab. **Students arriving later than 5 minutes to lab will not be admitted. Once again, reasonable exceptions may be made.**

7. **Write Up:** The conclusion of a lab requires a short write up, the project will require a formal lab report (more on that later) that summarizes the in- lab work. A typical write up must have an introduction, summary of a results, discussion covering the work done in-lab (not the pre-lab), and a conclusion. See the template for details. Write ups due time will be posted on Canvas, typically due in one week, by 8:30am on the day of the of next lab.

Consider the following when writing your lab write up.

- For every value listed, include the appropriate units.
- Be careful of the number of significant digits in the calculated results. This number should not be greater than the least number of significant digits in the data used for the calculations.
- Be objective and do not use colloquial phrases.
- Always give numerical results when applicable. Do not just say, “the results were as expected.” Specify the actual results, the expected results, any errors, and explain them.
- Be concise. A short direct write up is more desirable than a long rambling one. But at the same time, don’t give one word answers. The reader should clearly be able to tell that the writer understands the subject material.
- Keep everything neat and label everything clearly.
- Remember to number the equations and use figure numbers and table numbers for any drawings or tables. Also include captions for all figures and tables.
- Percent error is defined as $\delta = \left| \frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} \right| \times 100\%$
- Use the file "Circuits 1 prelab and write up template" in the "Lab related files" folder as a template for both Pre-Lab and Write-up reports.
- The write up must be typed, including tabled values, no pictures.
- All figures and tables must have a number and a caption.
- Include section labels, if it’s unclear which section is which, you’re more likely to lose points for not having a section.
- Don’t make large unreadable tables and avoid weird table placement. All values must be appropriately tabled.
- Don’t use large spacing, single spaced is fine.
- Don’t include figures from the lab manual or equations unless they’re asked for. You also don’t need to show how you calculated voltage or percent error. Percent error is an absolute value.

- Avoid rephrasing or copying the objective from the lab manual. The introduction and conclusion are worth a lot in the write up because the expectation is that you'll put time into them. There is a cap of four sentences but that doesn't mean you can get away with one or two weak sentences.
- The discussion section should be a discussion on the results of lab and not discussing how you did the lab. What do the results show? You should aim for a few paragraphs.

1.3 Tentative Schedule

Subject to change.

- Lab 1&2 - Week 2
- Lab 3 - Week 3
- Lab 4 - Week 4
- Lab 5 - Week 6
- Lab 6 - Week 7
- Lab 7 - Week 8
- Lab 8 - Week 10
- Lab Project - Week 11-12

1.4 Grading

The grade for the lab section is made up labs and a project. Each lab is composed of a pre-lab, in-lab, and a write up. Students must complete the pre-lab, online submission, and circuit demo, to complete the in-lab portion, failure to complete the pre- lab will result in a zero for the entire lab.

- Each lab is worth 10% and the project is worth 20% of the total lab grade. There is a total possible score of 110% for the lab. A minimum score of 70% in lab is required to pass the course, subject to change based on overall student performance. Completing the project is also required to pass the course.
- There is an on-line prelab quiz in Canvas, due at the same time with the prelab report (8:30am of the lab day). It is a few basic concepts required for the lab and worth 10 points (except Lab 1). Prelab report is worth 40 points, including the 10 points prelab demo circuits if required. Students are required to attend the lab to complete their in-lab items to get 20 points. Incomplete lab will get zero points for the In-lab. No partial points. In-lab results are required to be checked for completion. Absence will result in zero points for both In-

Lab (20 points) and Write-Up (30 points), because there is no proof of individual work performed.

- Each individual lab is graded as follows and will be accompanied by a rubric when appropriate:
 1. Pre-lab - 50% = 10% pre-lab quiz + 40% pre-lab report (this includes the demo points)
 2. In-Lab - 20% You either get these points or you don't. There is no partial credit
 3. Write Up - 30% If absent, this is automatically a 0, since there is no proof of personal work

1.5 Makeup Policy

Students are only allowed to make up missed labs with a legitimate absence and must produce documentation for their absence.

1.6 Copying

Students are bound by the University of Florida honor code and every student is expected to produce their own work. Any students who are found copying will receive a zero for the lab.

1.7 Safety

Although unlikely, it is possible to become injured in this laboratory. Emergency contacts by phone are 911 and 395-0050, the latter is for the Emergency Room at Shands Hospital. On some university phones it may be necessary to dial 9 first.

When working in any electrical laboratory, always keep electrical safety in mind. The following are some safety rules that each student should be aware of before beginning a laboratory experiment. Some of the rules do not apply to this laboratory, but to other electrical laboratories.

1. Never work totally alone in the laboratory. Someone else, and preferably the laboratory TA, should be present in the case of an emergency.
2. Use only the equipment provided. Do not use other equipment unless the laboratory TA approves the use.
3. Turn off the power before handling any wires. Never use damaged items, whether they are leads, components, equipment, or any other item.

4. To decrease the chances of being shocked, wear dry shoes and do not stand on metal or wet concrete. Also, do not wear any metal or jewelry. Moreover, do not handle wires, components, or equipment with wet hands.
5. In a laboratory in which soldering irons are used, keep an attentive eye on a hot soldering iron. Also, place it in the proper holder when not in use. Never leave a hot soldering iron unattended.
6. Make no connection to the power supply until the very last step. This practice will ensure that a student who is handling leads will not be shocked. Also, the circuit being built will not be harmed during construction.
7. Should someone become incapacitated because of electric shock, no one should touch that person until the power is turned off. Otherwise, the rescuer could also be shocked. Call for emergency services as soon as possible since resuscitation is likely if treatment is applied quickly. If breathing has stopped, begin CPR immediately and continue until qualified medical assistance has arrived.
8. If a piece of equipment is not working properly, attach a note and report the problem to the lab TA. The malfunction could possibly lead to a life-threatening situation.

Chapter 2

Lab 2 - Introduction to the Digilent Analog Discovery

2.1 Objective

The purpose of this laboratory is to familiarize students with the Digilent Analog Discovery and its associated software suite.

2.2 Materials

- Laptop with Waveforms required
- Digilent Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit
- Multimeter

2.3 Introduction

The Analog Discovery combines a variety of traditional test bench tools into a simple embedded device that fits in the palm of your hand. This lab focuses on bringing students up to speed on how to use the variety of tools associated with the Analog Discovery.

There are a variety of tutorials and how-to examples on getting started with the Analog Discovery and its associated software. To download Waveforms and see more information, read the [Digilent Getting Started Guide](#).

There are also a variety of tutorials for working with the specific tools:

- [Using the Waveform Generator](#)
- [Using the Oscilloscope](#)
- [Using the Power Supplies](#)

Digilent also has the same tutorials as videos:

- Analog Discovery 2 Quick-Start: Video1 - [Unboxing and Software Download](#)
- Analog Discovery 2 Quick-Start: Video 2a - [Installing WaveForms on Windows](#)
- Analog Discovery 2 Quick-Start: Video 2b - [Installing WaveForms on Mac](#)
- Analog Discovery 2 Quick-Start: Video 2c - [Installing WaveForms on Linux](#)
- Analog Discovery 2 Quick-Start: Video 4 - [Oscilloscope Tool](#)
- Analog Discovery 2 Quick-Start: Video 5 - [Waveform Generator Tool](#)
- Analog Discovery 2 Quick-Start: Video 9 - [Power Supplies](#)

Additionally, a previous student, David Munzer, has created a few tutorials in power point that can be accessed through Canvas.

- Analog Discovery Waveforms Tutorial
- Breadboarding Tutorial
- Debugging Tutorial

2.4 Big Picture

This portion of the lab usually shows how the circuits in this lab fit in to the final project but for the introductory labs, it's left out.

2.5 Pre-Lab Requirements

To be admitted to the lab, you must have an Analog Discovery board, breadboard, wiring kit, and a laptop with Waveforms 2015 (or later) installed. Before your first lab session, connect your Analog Discovery board to your laptop to verify that the device is recognized by Waveforms. If you encounter any issues, notify your lab instructor immediately.

Work through the various tutorials to get a feel for the Analog Discovery, Waveforms software, and breadboarding. Use wires from a solderless breadboard wiring kit as pictured below, the kit will be provided as part of your lab material.

There will also be a soldering activity to prepare students for their final project, where they will

solder components onto a printed circuit board (PCB). In this activity, students will have to solder an integrated chip (IC) onto practice boards. Read [“Learn how to Solder in 5 mins”](#) and watch [“Soldering Crash Course”](#) before entering your lab section to prepare for this activity.

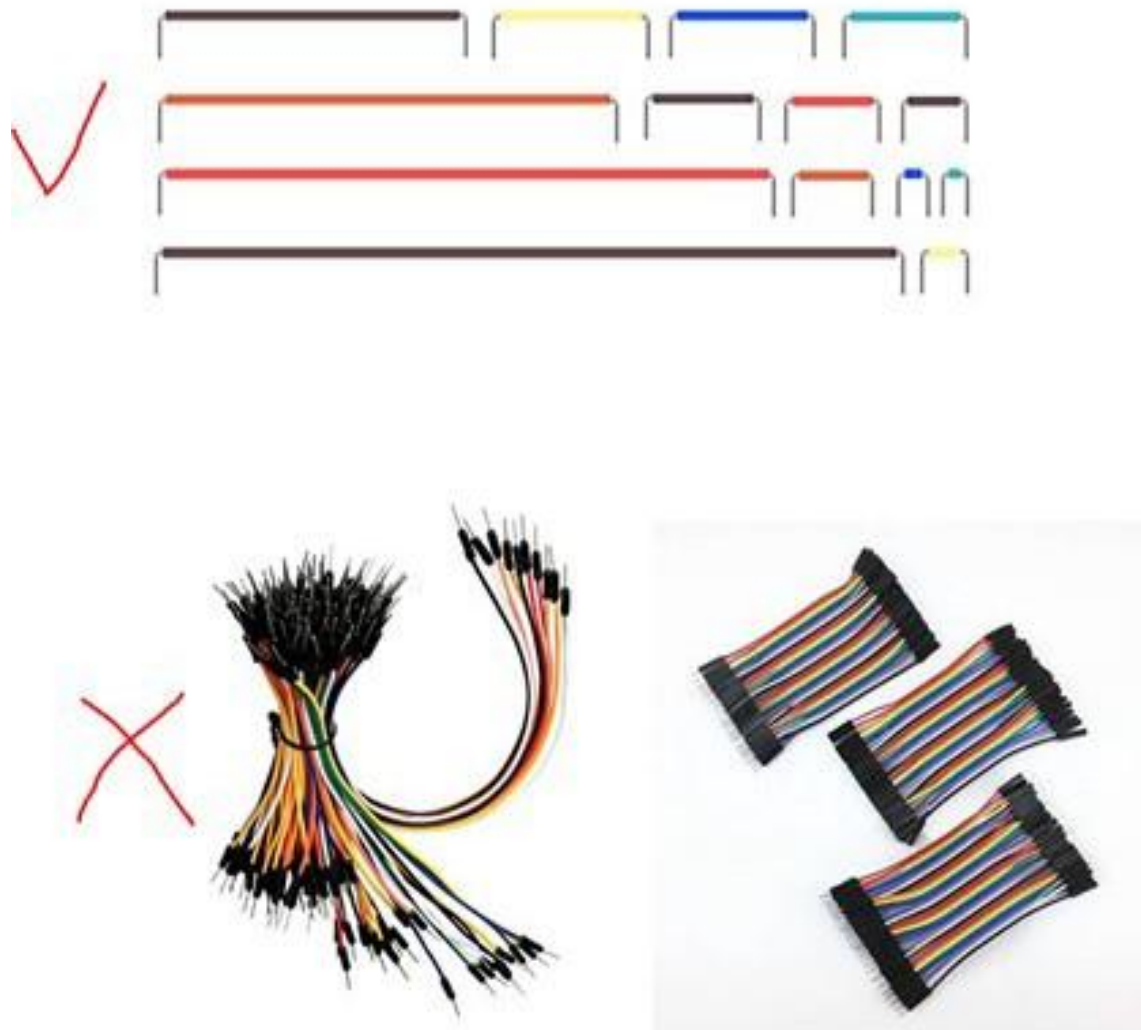


Figure 2.1: Example of the wires you should use.

2.6 In-Lab Requirements

To be accomplished in lab:

1. Pre-lab check: laptop should have Waveform installed and connected to the DAD

2. Review of Lab Rules and Policies
3. Live tutorial
4. Soldering demonstration

2.7 Write Up

This section would detail the requirements for the write up but there is no write up for this lab.

Chapter 3

Lab 3 - LTspice

3.1 Objective

The objective of the laboratory is to introduce students to SPICE (Simulation Program with Integrated Circuit Emphasis) and illustrate the differences between real and ideal components.

3.2 Materials

- Laptop with LTspice

3.3 Introduction

For the purposes of this class, LTspice (current Windows version is 24.1.0) will be used. It is a free version of spice that's supported by Linear Technology (recently acquired by Analog Devices). It can be downloaded from Linear Technologies' website [here](#).

There are useful resources provided in the Lab 3 Canvas folder with Lab 3 tips and LTspice keyboard shortcuts.

If possible, use LTspice on Windows for the best experience. The Mac version looks bare bones compared to the Windows version, but all the same functionalities exist, it's just hidden in sub menus.

Below are some videos to help you get familiar with the software.

- LT - [LTspice IV Overview](#)
- LT - [Schematic Editor](#)
- LT - [Waveform Viewer](#)
- LT - [Stepping Parameters](#)

3.4 Pre-Lab Requirements

Install LTspice, work through the tutorials, and complete the following.

3.4.1 Modeling Simple Circuits

A resistor ladder is shown in Figure 3.1.

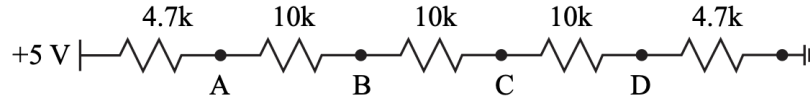


Figure 3.1: Resistor ladder example.

1. Using LTspice, layout the circuit in Figure 3.1 and run a DC operating point simulation (.op) to generate the node voltages at nodes A, B, C, and D. Table the node voltages.
2. When measuring the node voltages in real life, an oscilloscope is used. The scope probes have a finite input resistance of $1\text{ M}\Omega$. Re-simulate the node voltages with a $1\text{ M}\Omega$ resistor at nodes A through D, Figure 3.2 (a) through (d). There should be a total of 4 simulations with node voltages A through D for each simulation. Table the values.
3. Save the tabled values to submit to Canvas as part of the pre-lab.

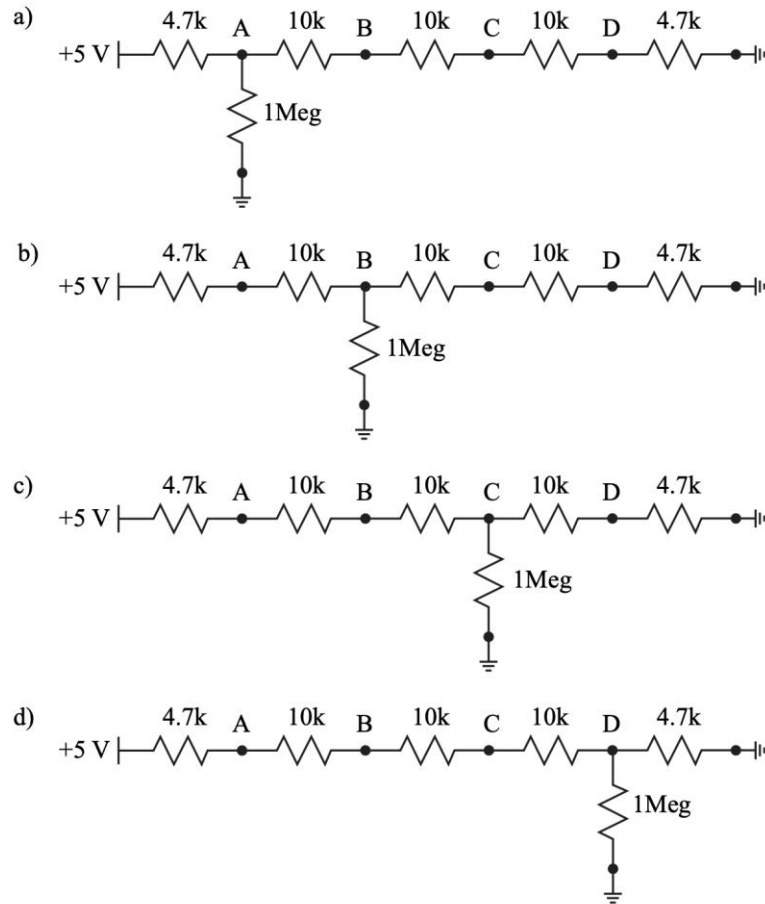


Figure 3.2: Resistor ladder with a 1 M Ω resistor attached at nodes A through D ((a) through (d)).

3.4.2 LTspice Variables

There is clearly an effect caused by the input resistance of the oscilloscope, often called the “loading effect”. Using LTspice, quantify the effect as a function of resistance by completing the following:

1. Layout the circuits in Figure 3.3 in LTspice on the same schematic. Use the variable {R} for resistor values, set the source voltage to 5 V DC and label each output node.
2. Use the following spice directive in conjunction with a DC operating point simulation (.op) to sweep the resistors in both voltage dividers.

```
.step param R list 10 100 1k 10k 100k 1Meg 10Meg
```

3. Plot the voltages at V_{out1} and V_{out2} and set the x axis to logarithmic by right clicking on the x axis. Save the results into a document, include your circuit image and simulation results, which will be submitted to Canvas as part of the pre-lab.

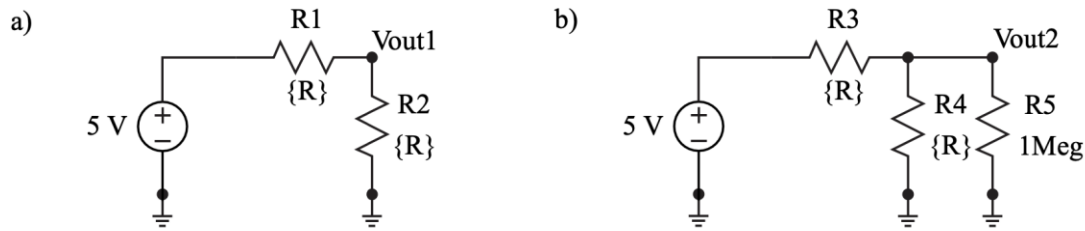


Figure 3.3: Two circuits to test the loading effect of one of the o-scope probes. The first circuit is a simple voltage divider that is not loaded (a) while the second circuit has a $1\text{ M}\Omega$ resistor in parallel with $R4$ (b).

3.4.3 Monte Carlo Simulations

Resistors have tolerances for their values which can vary from 10% for generic through hole resistors to as low as 0.005% for laser trimmed resistors. There is never a guarantee that a resistor has a resistance exactly equal to its value.

LTspice has a Monte Carlo function, $\{mc(x,y)\}$, that will randomize a number x with a tolerance of y between $x*(1+y)$ and $x*(1-y)$ with a uniform distribution.

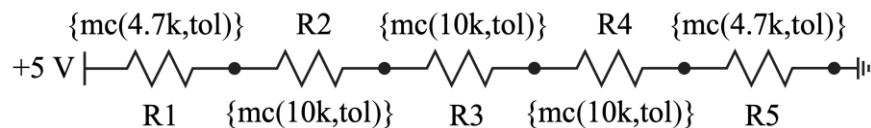


Figure 3.4: Resistor ladder with the values replaced by the LTspice Monte Carlo function.

Complete the following:

1. Layout the circuit in Figure 3.4, setting the resistor values to call the mc function.
2. Use the following spice directive in conjunction with a DC operating point simulation to randomize all the resistor values for 1000 trials with a 5 percent tolerance using the two directives below.

```
.param tol=0.05
.step param run 1 1k 1
```

3. Plot all the node voltages, A-D, on separate plot planes in one graph to view how the voltage at each node varies for each trial. Table the voltage ranges (V_{max} and V_{min}) for each node.
4. Save the results into a document, include your circuit image and simulation results, which will be submitted to Canvas as part of the pre-lab. Use " .meas " spice directive to measure

the voltage range (Vmax and Vmin) of nodes. To view the node voltages and currents, open the Spice Error Log under the View tab. Depending on your LTspice version, you may have to scroll all the way down to view the values (use the side scrolling bar).

However, if your LTspice “.meas” directive does not work (i.e., the maximum and minimum values display as the same), you will need to manually determine the values. To do this, hover your cursor over the plot panes in the simulation results and read the Y-axis values to identify the maximum and minimum points.

Below is an example of using the “.meas” directive to find the maximum voltage at node A:

.meas VAmix max V(A)

VAmix is the new variable, **max** is the command to find the max of **V(node name)**

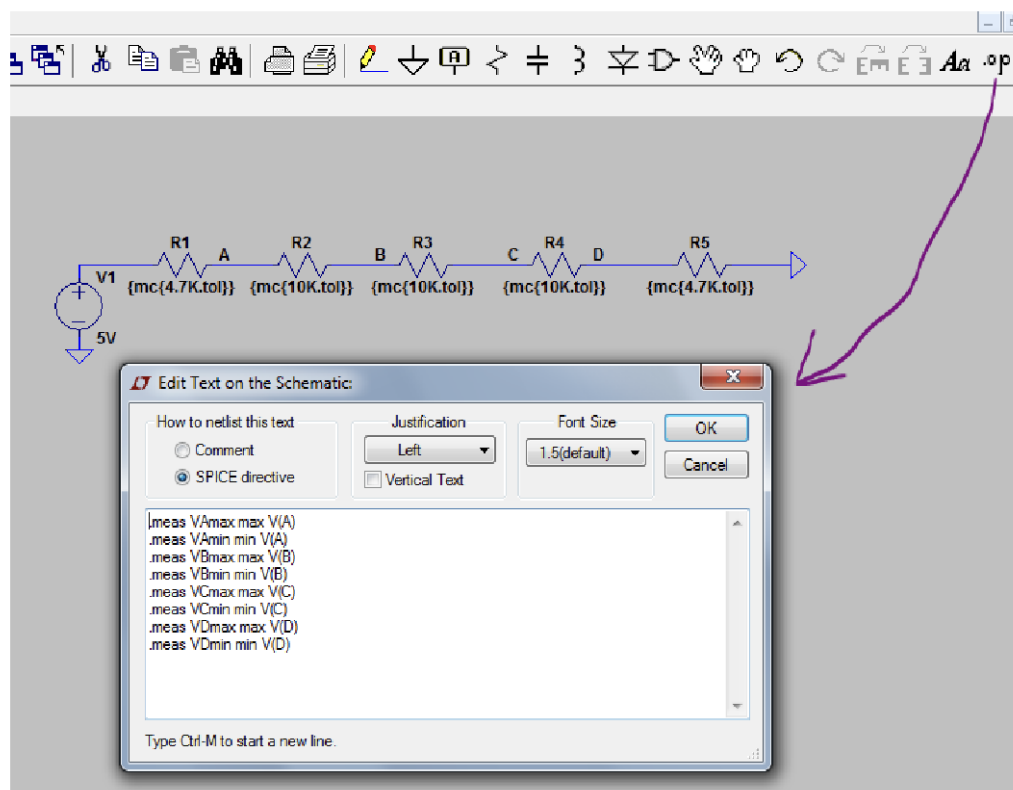


Figure 3.5: Example of using the max measure directive.

3.5 In-Lab Requirements

The following must be completed and submitted to Canvas before the start of lab.

1. Section 3.4.1 - Tabled voltages (note: four sets for 3.4.1.2).
2. Section 3.4.2 - Image of circuit and simulation results.
3. Section 3.4.3 - Image of circuit and tabled voltages and simulation results.

Complete the following items in lab.

3.5.1 Equivalent Resistance

An 8-bit R-2R ladder is a digital-to-analog converter (DAC) designed to convert an 8-bit binary input into a corresponding analog voltage. It uses a repeating pattern of resistors with values of R and $2R$ to create a voltage divider network. The binary input determines which switches in the ladder are activated, generating a proportional analog output. R-2R ladders are widely used in applications requiring accurate and efficient digital-to-analog conversion, such as audio devices and microcontroller-based systems.

When building the below circuit in spice, instead of assigning each resistor value to 470, give them the value "Ra" and create a spice directive ".param Ra=470". This will allow you to then change the directive's value from 470 to 500 afterwards and then rerun it to complete the experiment. This is a much more effective method.

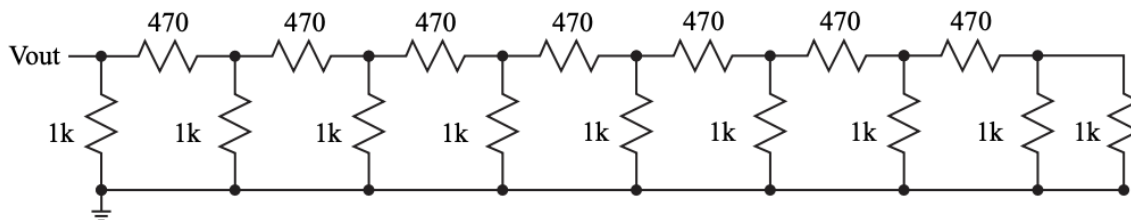


Figure 3.6: An 8-bit R-2R ladder with all the digital input bits tied to ground.

- a) Using a 1 V test source at V_{out} run a DC operating point simulation and calculate the equivalent resistance using Ohm's Law. Save the equivalent resistance value.
- b) Change 470 Ω resistors to 500 Ω resistors and confirm that the equivalent resistance is 500 Ω .

3.5.2 Monte Carlo with Different Tolerances

Repeat the Monte Carlo simulations from Section 3.4.3 using the following tolerances. Table the voltage ranges at each node.

1. 1%
2. 0.05%
3. 0.005%

3.5.3 Resistor Costs

While tighter resistor tolerances are desirable for improved accuracy, they often come at a higher cost. Using [DigiKey](#), determine the lowest total cost of constructing the 8-bit R-2R ladder shown in Figure 3.6, considering only the cost of resistors (500 Ω and 1 k Ω). Use the price listed for a quantity of 1 for each resistor. Search for Through Hole Resistors, in-stock, 0.25 Watt for the tolerances listed below: (If the item is out of stock, you may use another website such as [Mouser](#).)

1. 1%
2. 0.1%
3. 0.05%

3.6 Write Up

Include the following in the write up (use the report template file in Canvas).

1. Two equivalent resistor values from Subsection 3.5.1.
2. Table of voltage ranges from Subsection 3.5.2.
3. Table of cost ranges from Subsection 3.5.3.

Discuss the impact of resistor variations on the R-2R ladder, including their influence on node voltages and overall equivalent resistance accuracy. Additionally, evaluate the trade-offs between using resistors with tighter tolerances and their associated costs. Are there other characteristics that should be considered when buying components? If yes, list and discuss them.

Chapter 4

Lab 4 - Resistive Circuits

4.1 Objective

The objective of the laboratory is to introduce basic resistive circuits, the effect of resistors in series and parallel, applications of simple resistor ladders, and the effects of real world resistors.

4.2 Materials

- Laptop with Waveforms
- Digilent Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit

4.3 Introduction

Resistors are a fundamental part of all electronic circuits. Understanding how they work and how to apply them is a basic requirement for any electronics course.

4.3.1 Resistor Combinations

Resistors in series will always add together:

$$R_{Total} = R_1 + R_2 \dots R_N \quad (4.1)$$

while resistors in parallel combine as the addition of reciprocals:

$$\frac{1}{R_{Total}} = \frac{1}{R_1} + \frac{1}{R_2} \cdots \frac{1}{R_N} \quad (4.2)$$

Series resistances are straight forward, add one resistor to the other, but the parallel combination leads to different results based on the values of the resistors in parallel. For instance, a large resistor in parallel with a small resistor gives an equivalent resistance approximately equal to the small resistor. Similarly, two resistors of the same value in parallel gives an equivalent resistance equal to half of either resistor.

4.3.2 Voltage and Current Dividers

Resistors are commonly used to create voltage and current dividers, with the choice depending on the application. They are often employed to generate a smaller voltage from a larger voltage or to produce a smaller current from a larger current.

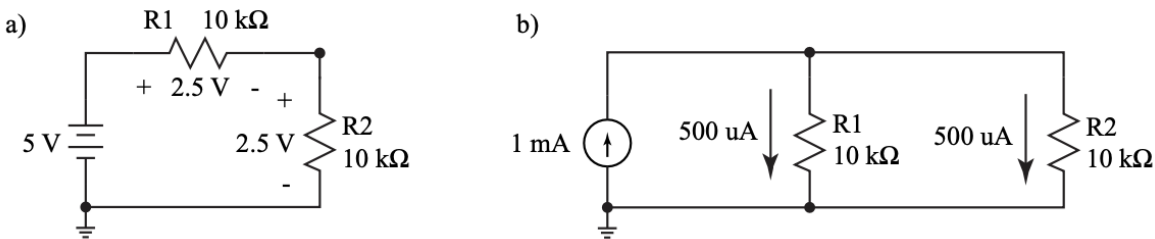


Figure 4.1: Examples of a voltage divider (a) and a current divider (b)

The output of a voltage divider can be simply calculated as:

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in} \quad (4.3)$$

In the example shown in (a), the output voltage is 2.5 V, which corresponds to the voltage across R2. This result is obtained by applying the voltage divider equation. Since the values for R1 and R2 are equal, the voltage drop across R1 is also 2.5 V.

$$\left(\frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} \right) 5 \text{ V} = 2.5 \text{ V} \quad (4.4)$$

Similarly, a current divider is calculated in the same way where:

$$I_{out} = \frac{R2}{R1 + R2} I_{in} \quad (4.5)$$

and the resulting current in each resistor is 500 μA which is again found by plugging in to the equation. Because the values for R1 and R2 are the same, the current through R1 is also 500 μA .

$$\left(\frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} \right) 1 \text{ mA} = 500 \mu\text{A} \quad (4.6)$$

Note that a voltage divider applies to resistors in series while a current divider applies to resistors in parallel.

4.3.3 Resistor Ladders

Resistors are often used in a string or ladder configuration to generate reference voltages with a specific spacing. Typical applications of a resistor ladder are analog to digital conversion where the input voltage is compared to voltages on the resistor ladder or digital to analog conversion where digital bits (1s and 0s) are converted to an analog voltage.

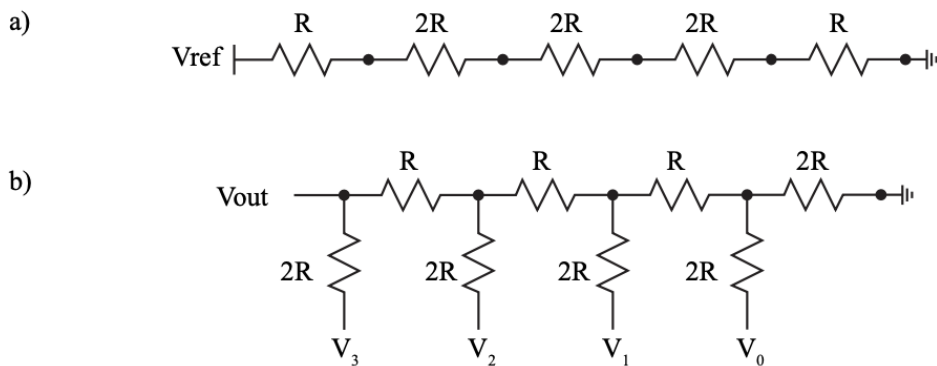


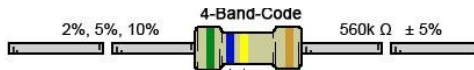
Figure 4.2: A simple resistor ladder where the resistor values set the voltages at each node (a) and a 4-bit R-2R ladder where digital bits, represented as voltages V_0 through V_3 are converted to an analog voltage, V_{out} .

4.3.4 Measuring Voltage and Current

In the past, separate meters were required to measure voltages and currents. Today, however, analog meters have been replaced by digital multimeters (DMMs). A typical DMM can perform various measurements, including AC and DC voltage, AC and DC current, resistance, and continuity. For this experiment, the focus will be on measuring DC voltage and current.

4.3.5 Resistor Color Code

There are a variety of resistor in the course kit and while it's possible to measure all the values individually, learning how to read the color codes will make identifying individual resistors much easier. See the look up table below.



COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

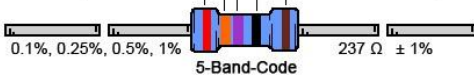


Figure 4.3: Resistor color code look up table.

4.4 Big Picture

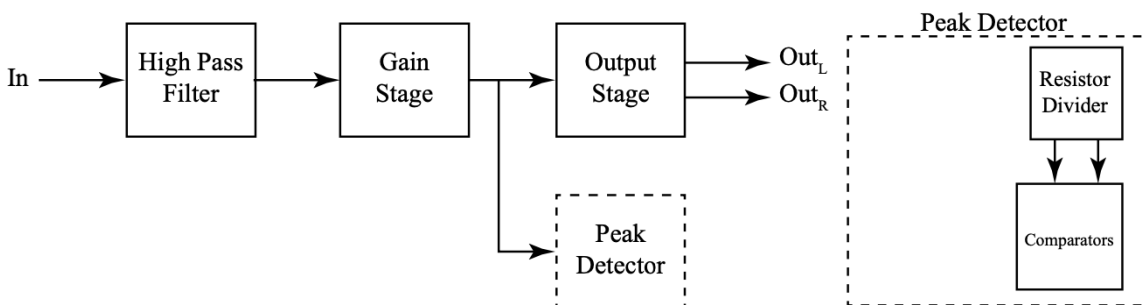


Figure 4.4: Big picture with emphasis on the peak detector.

This lab focuses on basic resistor circuits which will help to form the resistor divider used to set the threshold voltages in the peak detector for the final project.

4.5 Pre-lab Requirements

Complete the following before coming to lab.

4.5.1 Resistor Ladders

1. Using the circuit in Figure 4.5, calculate the ideal voltages for nodes A, B, C, and D. Table the values for future comparison.
2. Construct the circuit in Figure 4.5 and measure the actual voltages using the oscilloscope. Table the values.

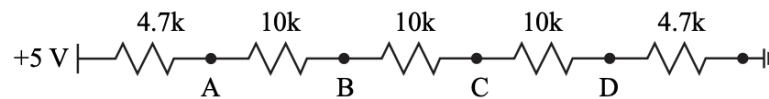


Figure 4.5: Resistor ladder example

4.5.2 Two-bit R-2R Ladder

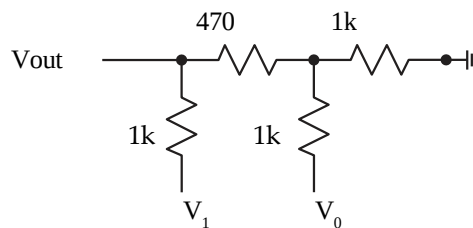


Figure 4.6: Two-bit R-2R ladder.

1. Using the circuit in Figure 4.6, calculate the ideal output voltages for the following combination of voltages and table the values:
 - (a) $V_0 = 0\text{ V}$ and $V_1 = 0\text{ V}$
 - (b) $V_0 = 5\text{ V}$ and $V_1 = 0\text{ V}$
 - (c) $V_0 = 0\text{ V}$ and $V_1 = 5\text{ V}$
 - (d) $V_0 = 5\text{ V}$ and $V_1 = 5\text{ V}$
2. Construct the circuit in Figure 4.6 and measure the output voltage using the oscilloscope and table the values. To generate the changes in voltages, set:

- (a) Wavegen 1 (V_0) to a 1 Hz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - (b) Wavegen 2 (V_1) to a 500 mHz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - (c) Make sure the Wavegen signals are set to synchronized.
3. Save the circuit and Wavegen settings to demo to your lab instructor.
 - (a) The output should look like the Scope image in Figure 4.7. In this circuit, V_0 and V_1 serve as digital inputs, switching between on (5 V) and off (0 V). As you “count” from 0 to 3 in binary, the resistor network creates proportional voltage steps at the output. This means that the output voltage represents the binary input as an analog voltage level between ground and 5 V, with each step increasing based on the weighted contributions of V_0 and V_1 .

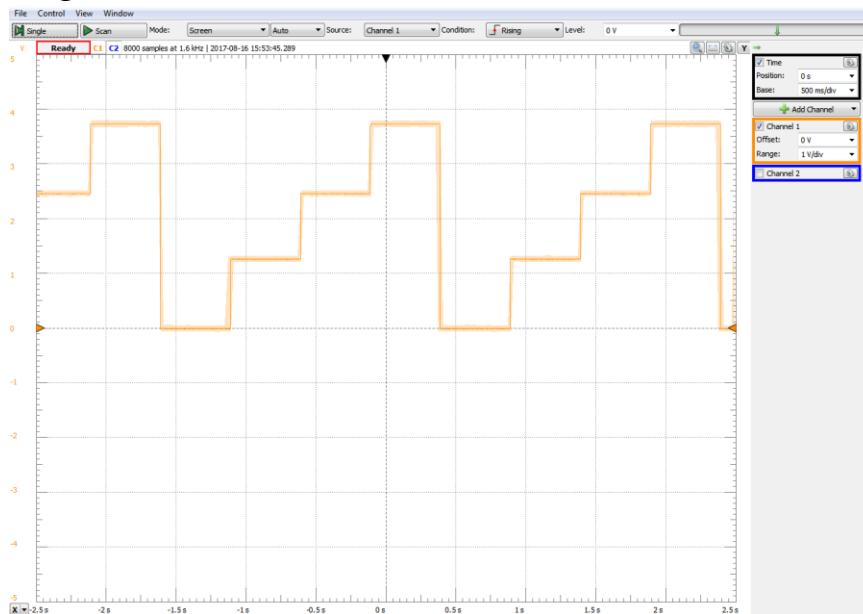


Figure 4.7: The desired output for the R-2R ladder circuit with the appropriate inputs.

4.6 In-Lab Requirements

The following items must be ready at the start of lab.

- Submit the table of voltages for the resistor ladder and the R-2R ladder to Canvas in a document.
- Demo the output of the R-2R ladder as V_0 and V_1 change.

4.6.1 Resistor Tolerances

There is no such thing as a perfect resistor. All resistors have tolerance about their specified value. For example, a $10\text{k}\ \Omega$ resistor with a $\pm 5\%$ tolerance will have a value between $10\text{k}\ \Omega - 5\% \leq R \leq 10\text{k}\ \Omega + 5\%$.

- Using the multimeter, measure the resistance of the resistors in the ladder from Subsection 4.5.1. Remember to remove the resistors from the breadboard so that they can be measured independently. Table these values.
- Re-calculate the ideal voltages using the measured resistances from the previous step and table the values.
- Using the multimeter, measure the resistance of the resistors used in the R-2R ladder from Subsection 4.5.2. Remember to remove the resistors from the breadboard so that they can be measured independently.
- Re-calculate the ideal voltages using the measured resistances from the previous step and table the values.

4.6.2 Three-bit R-2R ladder

Find someone in the lab to form a team of two. If there is an odd number of students, allow for a team of three where necessary. Complete the following:

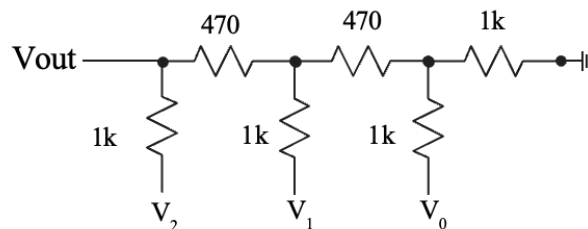


Figure 4.8: Three-bit R-2R ladder.

1. Construct the circuit in Figure 4.8 and measure the output voltage using the oscilloscope and table the values. To generate the changes in voltages, set:
 - (a) Wavegen 1 (V_0) to a 1 Hz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - (b) Wavegen 2 (V_1) to a 500 mHz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.

- (c) Make sure the Wavegen signals are set to synchronized.
 - (d) Since there is no third Wavegen wire, you will use the DAD's V+ wire (red) and ground wire (black) for V_2 .
2. Take a screenshot of the waveform of V_{out} , measure the values of each step, and record them in a table for the following cases:
- (a) $V_2 = 0\text{ V}$.
 - (b) $V_2 = 5\text{ V}$.
3. Answer the questions listed in the Write up Section. Include the name of your teammate(s).

4.7 Write Up

Detail the effects of the resistance tolerances on the two different resistor ladders. Include the tabled values for both circuits and calculate the percent error using the experimentally measured values from the prelab and the values calculated from the measured resistors. You should have two tables: one for voltage values and their percent errors, and another for the resistor values and their percent errors.

Discuss the results of the teamwork section and answer the following:

- What did you learn from the teamwork section in terms of communication and collaboration?
- How would you rate your team's performance using a scale of 1 to 5, with 5 being the best? Discuss any errors your team might have had and how you corrected them.
- How is a three-bit ladder different than the two-bit ladder constructed in the prelab? What are some possible applications for the three-bit ladder?

For all lab write-up submissions and reports, the backgrounds in LTspice and Waveforms should be changed from dark to light to ensure readability for grading.

If the prelab simulation is incorrect, do not compare the in-lab results to wrong values. Correct numbers or circuit schematics should be used in the write-up report.

Chapter 5

Lab 5 - Operational Amplifiers

5.1 Objective

The purpose of this lab is to introduce operational amplifiers through the ideal op amp model, spice op amp model, and practical applications of op amps in common configurations.

5.2 Materials

- Laptop with LTspice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit with TLV272

5.3 Introduction

There is a large variety of op amp and specialized amplifiers. For this lab the more recent TLV272 is used, a dual op amp, which contains two op amps in a single chip.

5.3.1 The Ideal Op Amp

Recall that an ideal op amp will have the following properties.

- Zero input current, often called the input bias current and represented as I_B or I_{IB} .
- Zero input offset voltage, represented as either V_{IO} or V_{OS} .
- Infinite input resistance, represented as R_{in} or r_i .

- Zero output resistance, represented as R_{out} or r_o .
- Infinite gain, represented as A or α and often called the open loop gain.

These quantities are represented in Figure 5.1.

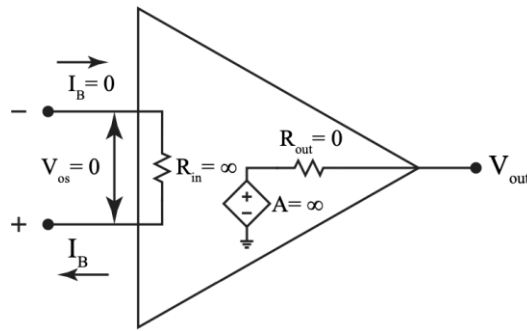


Figure 5.1: Basic op amp model.

However, as should be expected, practical op amps do not have these ideal quantities. For example, Texas Instrument's TLV272 has finite values for all the ideal properties. Note that these values vary and depend on production variation, temperature, and supply voltage.

- Finite input bias current, typically 1 pA.
- Finite input offset voltage, typically 0.5 mV.
- Finite input resistance, typically 1,000 G Ω differential input resistance.
- Finite gain, typically 115 dB (563k)

5.3.2 Common Configurations

There are a variety of configurations op amps can be used for but there are a few common configurations that form the basis of more complicated op amp circuits.

Inverting Amplifier

An inverting configuration, Figure 5.2 (a), takes its name from the inversion that happens from input to output, giving a output voltage proportional to R_2 and R_1 independent of the actual op amp properties.

$$V_{out} = -\frac{R_2}{R_1}V_{in} \quad (5.1)$$

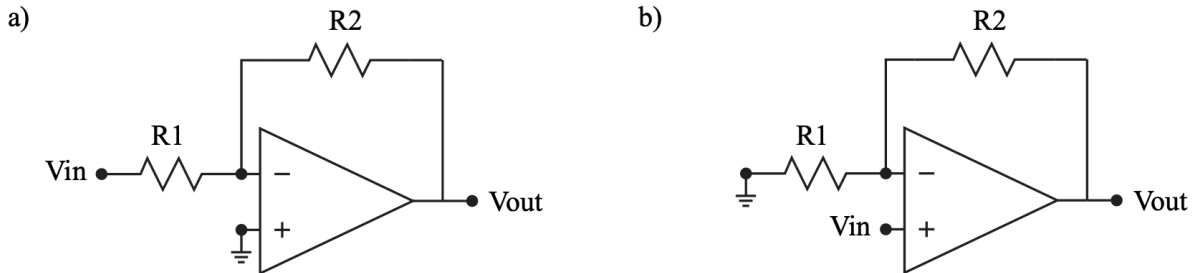


Figure 5.2: Different op amp configurations: inverting amplifier (a) and non-inverting amplifier (b)

Non-Inverting Amplifier

Similar to the inverting case, Figure 5.2 (b), this configuration results in an output voltage proportional to R_2 and R_1 but without an inversion and always has a gain of at least 1.

$$V_{out} = \left(1 + \frac{R_2}{R_1}\right)V_{in} \quad (5.2)$$

There is also a special case of the non-inverting amplifier where it is connected as a voltage follower or unity gain amplifier as show in Figure 5.3.

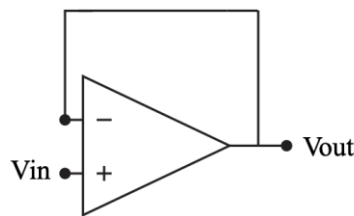


Figure 5.3: A non-inverting amplifier configured as a unity gain buffer.

The output is related directly to the input where:

$$V_{out} = V_{in} \quad (5.3)$$

and while this configuration may seem redundant, it has the advantage of providing a large input impedance to the source (V_{in}), minimizing loading effects, and a small output impedance to the load (V_{out}), ensuring efficient signal transfer.

5.3.3 Op Amp Pins

A pinout is a connection diagram for an integrated circuit, showing how its pins connect to the internal op amp terminals. It should not be confused with the pins in a spice simulation. Op amps typically have a standardized pinout, allowing chips to be easily swapped out. Figure 5.4 shows the pinout for the TLV272 with the op amp diagram drawn within.

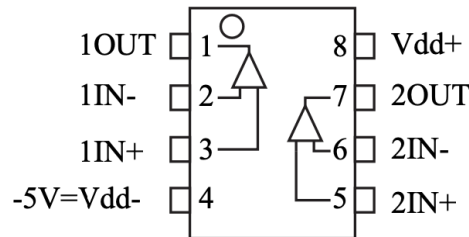


Figure 5.4: Pinout for TI's TLV272. Note that two op amps are contained, often called a dual op amp, within a single chip.

Name	Number	Description
1OUT	1	Output of the first op amp
1IN-	2	Inverting input of the first op amp
1IN+	3	Non-inverting input of the first op amp
Vdd-	4	Negative supply
2IN+	5	Non-inverting input of the second op amp
2IN-	6	Inverting input of the second op amp
2OUT	7	Output of the second op amp
Vdd+	8	Positive supply

Table 5.1: Pin descriptions from the TLV272 datasheet.

The pins are numbered from 1 to 8, counting counterclockwise starting in the upper left. Descriptions of each pin are provided in Table 5.1. Most of the pins should be familiar from class.

Polarity, or the orientation of the chip, is determined by a circle or indentation on the chip itself. The TLV272 has a small circle indentation in the upper left hand corner to indicate polarity and the position of pin 1.

The LTspice pinout differs from the physical chip's pinout, which is specified in the subcircuit .LIB file. Table 5.2 provides the corresponding pin descriptions for reference.

Number	Description
1	Non-inverting input
2	Inverting input
3	Positive supply
4	Negative supply
5	Output

Table 5.2: Pin descriptions from the TLV272 .LIB LTspice file.

5.4 Big Picture

This lab focuses on op amps which will be used in the gain and outputs stages of the final project.

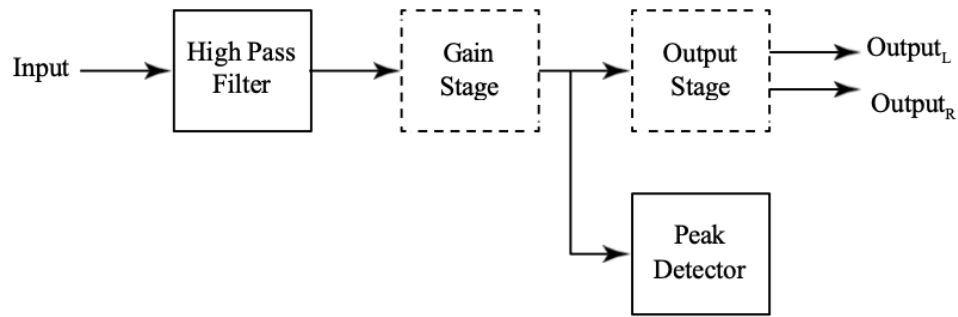


Figure 5.5: Big picture with emphasis on the gain and output stages.

5.5 Pre-Lab Requirements

Complete the following before coming to lab.

5.5.1 LTspice TLV272 Op Amp Model

Complete the following using the TLV271 spice model, always power the op amp with ± 5 V for V_{dd+} and V_{dd-} .

1. Obtain the [TLV2 spice model](#) from TI's website. Note that this is the model for the TLV271 which is the same op amp that's duplicated in the TLV272. If you have trouble accessing the zip file using the link, the file can also be accessed from the lab 5 folder on canvas.
2. Import the model in to LTspice using the "LTspice Component Import" documents in the canvas lab folder. Video instructions can be found [here](#). The pin numbers in the spice model correspond to the numbers in Table 5.2, not the numbers in Table 5.1.

3. Familiarize yourself with the TLV272 [datasheet](#). Make note that pin 4 should be connected to -5 V, NOT ground. It is also recommended to use the schematic in Figure 5.4 rather than the schematic in the datasheet to avoid confusion when constructing your physical circuits.
4. Using the voltage follower configuration in Figure 5.3, run a DC operating point simulation (.op) and determine the V_{out} when V_{in} is zero volts; effectively calculating the offset voltage, table the value.
5. With the same voltage follower configuration, set V_{in} to a 1 V amplitude pulse. Vinitial(V): 0, Von(V): 1, Tdelay(s): 0.5m, Ton(s): 10s, Ncycle: 100. Leave any remaining variables blank.

Run a transient analysis with a stop time of 1m (.tran 1m). Plot the input and output voltage from 495 us to 505 us and then find the time it takes for the output to reach 1 V; the time should be on the order of micro seconds. This is a measure of slew rate in V/us, table the value. Save an image of the circuit and the plot of the input and output voltage for submission to canvas.

6. Configure an inverting amplifier, Figure 5.2 (a), with a gain of 10 by setting $R_2 = 10k$ and $R_1 = 1k$. Set the input voltage to a sine wave with the following parameters, DC offset(V): 0, Amp(V): 1, Freq(Hz): 1k, and the remaining variables blank.

Run a transient analysis with a stop time of 1m (.tran 1m). Plot the input and output voltage and record the max and min values of V_{out} , this is a measure of the maximum output swing, table the value. Save an image of the circuit and the plot of the input and output voltage for submission to canvas.

7. Using the same inverting configuration, set V_{in} to a 0.1 V DC source and run a transient analysis with a stop time of 1m (.tran 1m). Plot $-V(vout)/V(vin)$, this is a measure of the effective gain, table the result. Save an image of the circuit and the plot of effective gain for submission to canvas.

5.5.2 Op Amp Construction

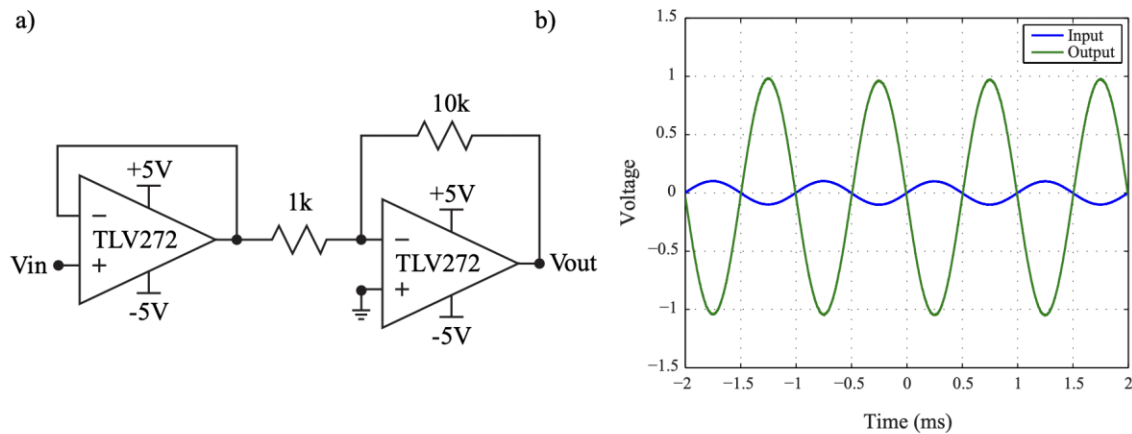


Figure 5.6: Schematic for the demo circuit, a voltage follower followed by an inverting amplifier with a gain of 10 (a) and the resulting output (b)

1. Using only one of the TLV272 op amp in your lab parts kit, build the circuit in Figure 5.6 (a) on a breadboard. Set the voltage supplies to ± 5 V and scope channel 1 to V_{in} and scope channel 2 to V_{out} . Configure the Wavegen to supply a 1 kHz sine wave with a 0.1 V amplitude. Your waveform should look like Figure 5.6 (b).

The demo circuit in Figure 5.6 consists of a voltage follower and an inverting amplifier with a gain of -10. The voltage follower ensures high input impedance and low output impedance, preventing loading while maintaining $V_{out} = V_{in}$.

This buffered signal then feeds into the inverting amplifier, where the gain is set by $V_{out} = -10V_{in}$, resulting in an amplified and inverted output. As shown in Figure 5.6 (b), a sinusoidal input produces a larger, inverted sinusoid.

5.5.3 Pre-lab Checklist

The following must be completed at the start of lab.

1. All the theoretical parameters from Subsection 5.5.1 tabled:
 - (a) Offset voltage.
 - (b) Slew rate.
 - (c) Maximum output swing (min and max output voltages).
 - (d) Effective gain.
2. Images of the output and circuits from Subsection 5.5.1 tabled:
 - (a) Item 5: Image of the circuit and the plot of the input and output voltage.

- (b) Item 6: Image of the circuit and the plot of the input and output voltage.
 - (c) Item 7: Image of the circuit and the plot of effective gain.
3. Op amp configurations from Subsection 5.5.2 built on a breadboard and working.

5.6 In-Lab Requirements

Complete the following during lab.

5.6.1 Experimental Op Amp Measurements

Using the pre-built op amp configurations, complete the following.

1. Using the voltage follower, Figure 5.3, input a 0 V DC voltage from the Wavgen and record the offset voltage. Choose an appropriate voltage scale, the voltage is on the order of mV. Save a screenshot of the waveform scope.
2. Using the voltage follower again, Figure 5.3, input a 0.5 V amplitude square wave with a 0.5 V offset from the Wavegen. Set the appropriate trigger and determine the slew rate in V/us. Save a screenshot of the waveform scope.
3. Using the inverting amplifier, Figure 5.2 (a), input a 1k Hz sine wave with a 1 V amplitude from the Wavegen. Determine the maximum output voltage swing by finding the maximum and minimum voltages.
4. Using the inverting amplifier again, Figure 5.2 (a), input a 0.1 V DC from the Wavegen. Determine the effective gain by dividing the output and input voltages. Choose an appropriate voltage scale for the input and output voltages.
5. Table all of values calculated in this subsection.

5.7 Write Up

Include the following in the write up.

1. Table of experimental parameters.
2. Screenshots of scope output plots.
3. Percent error, using the prelab values as the theoretical and the in-lab values as experimental.

Discuss the differences between the ideal op amp model, spice TLV272 model, and the physical

op amp focusing on the quantities calculated for this lab: offset voltage, slew rate, output voltage range, and effective gain. Touch on the following in your discussion.

- When is it appropriate to use one model over the other?
- How accurate should the spice model for an op amp be?

For all lab write up submissions and reports, the backgrounds for LTspice and Waveforms should be changed from dark background to light background to make them readable for grading.

If the prelab simulation is incorrect, do not compare the in-lab results to wrong values. Correct numbers and circuit schematics should be used in the write-up report.

Chapter 6

Lab 6 - Operational Amplifiers Applications

6.1 Objective

The purpose of this lab is to delve into the applications of operational amplifiers. There is an endless number of applications, the possibilities are often not intuitively obvious for students seeing them for the first time.

6.2 Materials

- Laptop with LTSpice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit with TLV272

6.3 Introduction

There are several standard op amp configurations. While Lab 5 demonstrated the simple inverting and non-inverting cases, this lab expands the scope to include both a summing amplifier and a difference amplifier.

Summing Amplifier

A summing amplifier, Figure 6.1 (c), is effectively the same configuration as an inverting amplifier but has multiple inputs that allow for a weighted output voltage.

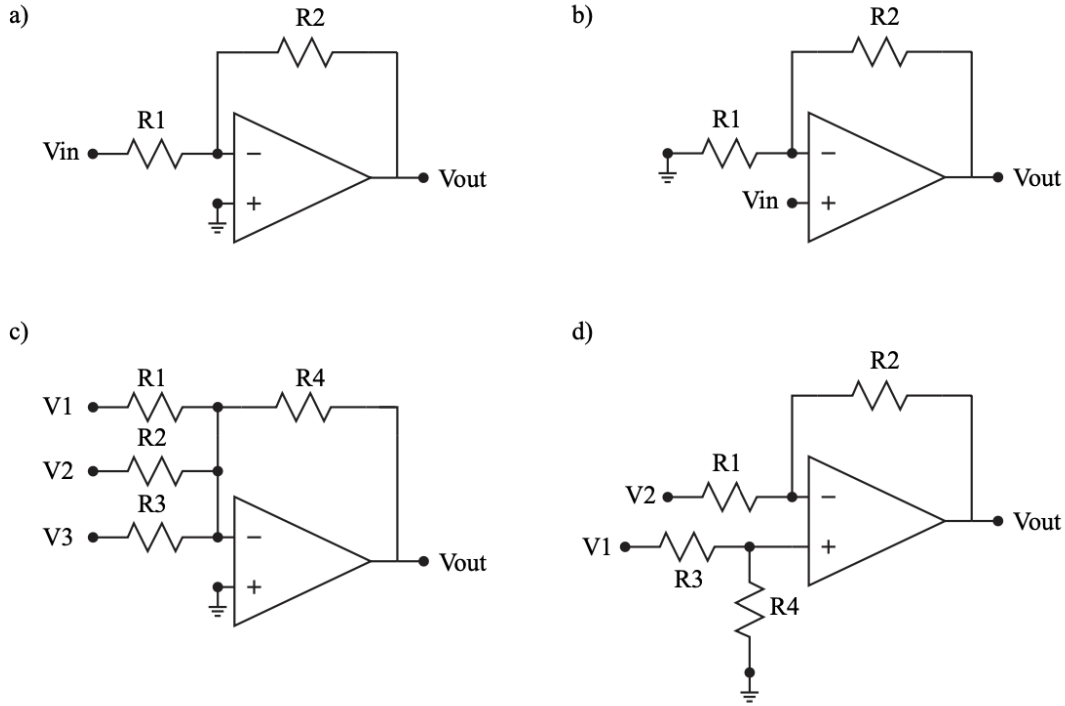


Figure 6.1: Different op amp configurations: inverting amplifier (a), non-inverting amplifier (b), summing amplifier (c), and difference amplifier (d).

$$V_{out} = -\left(\frac{R_4}{R_1}V_1 + \frac{R_4}{R_2}V_2 + \frac{R_4}{R_3}V_3\right) \quad (6.1)$$

When the resistors R_1 , R_2 , R_3 , and R_4 are equal, the output voltage is an equal combination of the input voltages. However, the resistors can be adjusted to assign different weights to the input voltages based on the application.

Difference Amplifier

Difference amplifiers, Figure 6.1 (d), takes the difference between two input voltages, with the output determined by the resistor value ratios.

$$V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{R_3 + R_4}{R_3} \right) - V_2 \frac{R_4}{R_3} \quad (6.2)$$

When the resistors $R_1 = R_3$ and $R_2 = R_4$ the output voltage simplifies to:

$$V_{out} = (V_1 - V_2) \frac{R_2}{R_1} \quad (6.3)$$

6.4 Big Picture

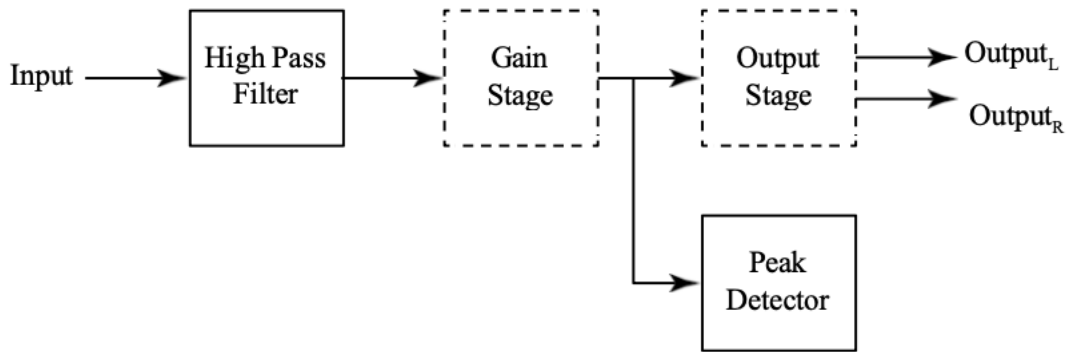


Figure 6.2: Big picture with emphasis on the gain and output stages.

This lab once again focuses on op amps, using them as building blocks to create more complicated circuits. The final project is about as complicated as the final circuit in this lab.

6.5 Pre-Lab Requirements

Complete the following before coming to lab.

For all op amp circuits, use net names for the voltage sources to keep schematics clean and easy to read. See Figure 6.3 for a visual.

Steps:

1. Place your voltage source in the schematic.
2. Instead of drawing wires directly from the source across the circuit, assign a net name (e.g., $V+$, $V-$, V_{in}).
3. Use the same net name wherever that connection is required in the schematic.

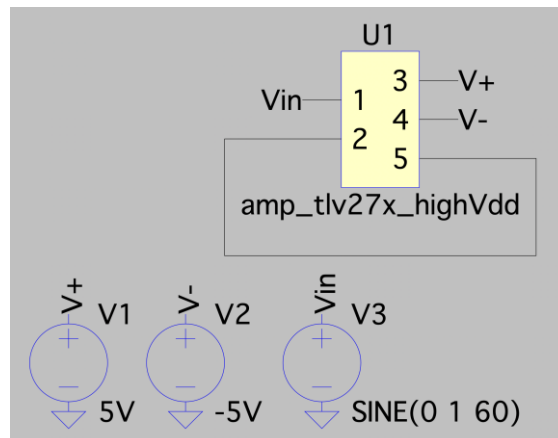


Figure 6.3: A voltage follower circuit to demonstrate the use of net names for voltage sources.

6.5.1 Buffering

Buffering is a key op amp application that addresses loading effects, which resistive circuits cannot easily solve. Op amp buffers provide high input impedance and low output impedance, preserving signal integrity across various applications.

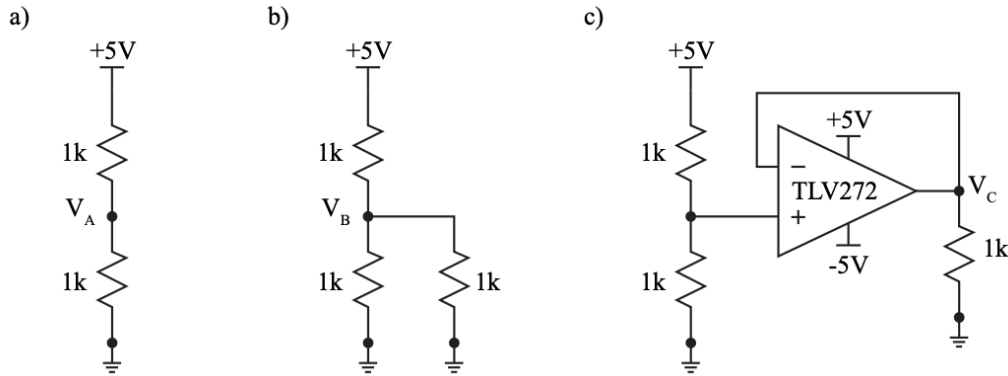


Figure 6.3: A simple voltage divider (a), voltage divider with a load (b), and a voltage follower between the output and the load (c).

1. Simulate all three circuits in Figure 6.3 using a DC operating point (.op). Keep all the circuits on the same schematic and table the voltages V_A , V_B , and V_C . Table your values and include images of your circuits.
2. Discuss why two of the voltages are approximately 2.5 V but one is not.

6.5.2 Level Shifting

Level shifting in electronic circuits is the process of changing the supply rails from one range to another. For example, when preparing a signal for analog to digital conversion, the signal may be bi-polar ($\pm$) and must be converted to a single voltage rail. So, instead of a sine wave that swings to +2.5V and -2.5V, 2.5V is added so that the sine wave swings from +5V to 0V with the same amplitude.

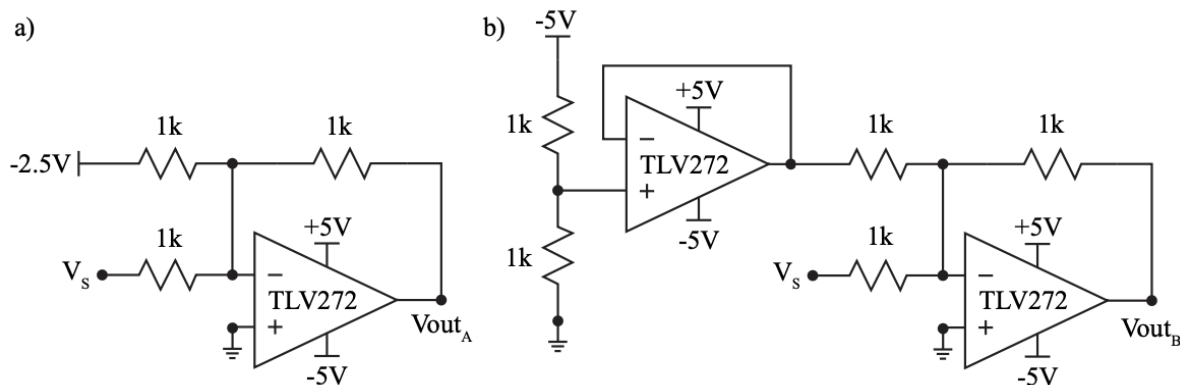


Figure 6.4: A summing amplifier used to add 2.5V to $-V_{IN}$ (a) and the same configuration where

the -2.5V source has been replaced with a buffered voltage divider (b).

1. Simulate the circuit in Figure 6.4 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.
2. Explain why the resistor divider in Figure 6.4 (b) uses -5 V instead of 5 V.
3. Practically, a 2.5 V reference may not be available for use and instead must be self-generated. Simulate the circuit in Figure 6.4 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.
4. Construct the circuit in Figure 6.4 (b) on a breadboard. Configure the Wavegen to generate a sine wave at 1 kHz with a 2 V amplitude and a 0 V offset. Plot the input and output on the Scope and save the circuit to demo at the start of lab. If you see clipping at your output, change the three 1k Ω resistors in the summing op amp to 33k Ω .
5. In the more general case, the full range of input must be accounted for. Simulate the circuit in Figure 6.5 using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN} to a sine wave with a 0 V DC offset, 5 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

This circuit uses 3 op amps to convert a ± 5 V sine wave into a +5/0 V sine wave. The top op amp generates a -2.5 V offset, while the bottom op amp buffers half of the input voltage. Both outputs are then fed into an (inverting) summing op amp to produce V_{outB} . In short, this circuit adds -2.5 V to half of a ± 5 V sine wave, then inverts it to obtain:

$$- [(-5/2) + (-2.5)] = 5 \text{ V and } - [(5/2) + (-2.5)] = 0 \text{ V}$$

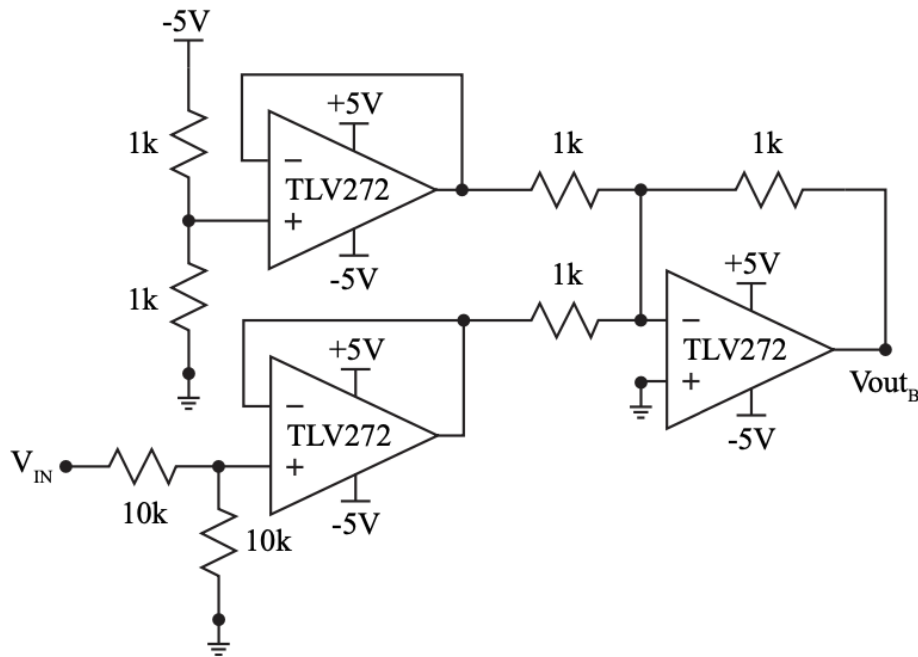


Figure 6.5: A DC level shifter configured to convert the full range of ± 5 V to +5/0 V.

6.5.3 Differential to Single Ended Conversion

Differential signals, which are two signals 180 degrees out of phase, offer improved noise performance because any interference affecting both signals in the same way cancels out when they are combined. In other words, only the difference between the signals is measured, removing unwanted noise. A difference amplifier can be used to convert the differential signal into a single-ended output.

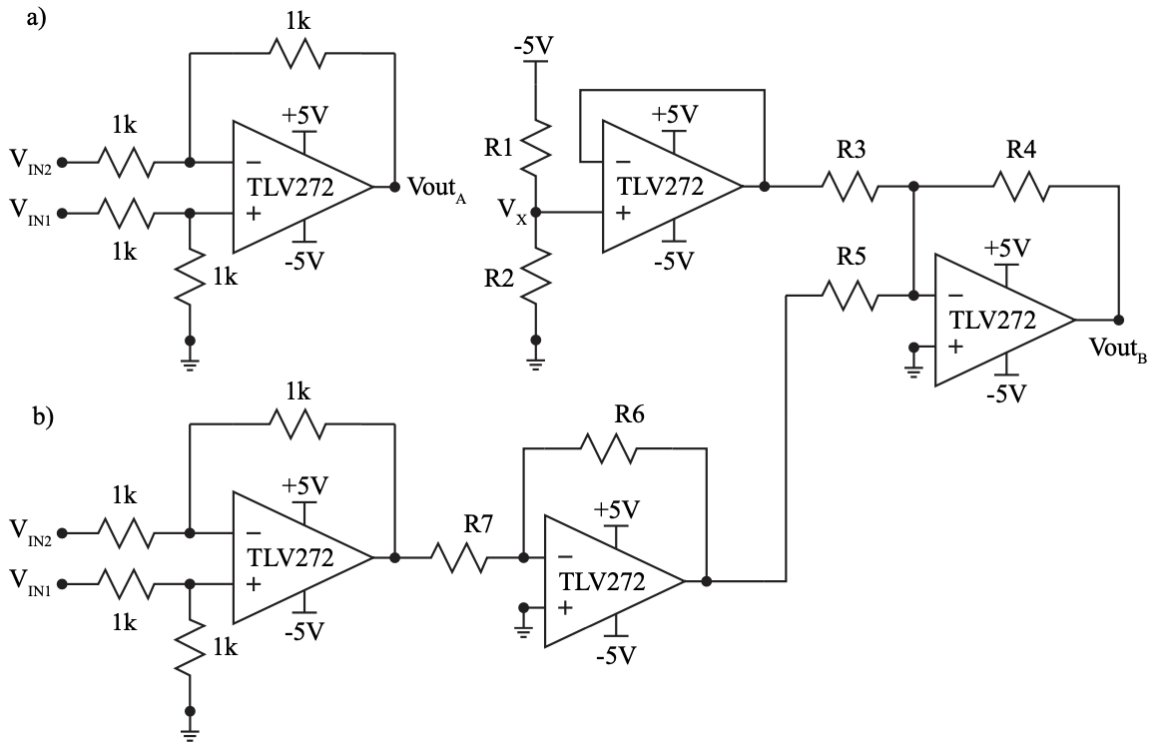


Figure 6.6: A difference amplifier that converts a differential signal to single ended (a) and a combination of the level shifter and differential to single ended converter (b).

1. Simulate the circuit in Figure 6.6 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN1} to a sine wave with a 0 V DC offset, 0.5 V amplitude, and a 1k Hz frequency. Set V_{IN2} to a sine wave with a 0 V DC offset, 0.5 V amplitude, a 1k Hz frequency, and a phase of 180 degrees. Save an image of the circuit and a plot of V_{IN1} , V_{IN2} , and V_{out_A} .
2. Determine the resistor values for R_1 through R_7 , recall that there are limited quantities in the lab kit. Choose R_1 and R_2 so that V_X is -2.5 V, R_3 , R_4 , and R_5 so that both voltages are added equally, and R_6 and R_7 so that the output, V_{out_B} utilizes as much of the full range, 0 to 5 V, as possible without clipping. You do not need to use resistors in parallel/series combinations to get an exact value.
3. Simulate the circuit in Figure 6.6 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{S1} and V_{S2} to their previously defined values. Save an image of the circuit and a plot of V_{IN1} , V_{IN2} , and V_{out_B} .
4. Explain how the circuit in Figure 6.6 (b) works in a few sentences, discussing the purpose of each op amp.

6.5.4 Pre-Lab Checklist

The following must be completed at the start of lab.

1. Section 6.5.1
 - (a) Item 1: Image of the circuit and a table of output voltages.
 - (b) Item 2: Answer to question.
2. Section 6.5.2
 - (a) Item 1: Image of the circuit and a plot of the input and output voltages.
 - (b) Item 2: Answer to question.
 - (c) Item 3: Image of the circuit and a plot of the input and output voltages.
 - (d) Item 4: Plot of the input and output voltages and a circuit demonstration.
 - (e) Item 5: Image of the circuit and a plot of the input and output voltages.
3. Section 6.5.3
 - (a) Item 1: Image of the circuit and a plot of the input and output voltages.
 - (b) Item 3: Image of the circuit and a plot of the input and output voltages.
 - (c) Item 4: Answer to question.

6.6 In-Lab Requirements

Complete the following in lab.

6.6.1 Construction

1. Construct the circuit from Section 6.5.2 Item 5. Set Wavegen 1, V_{IN} to a sine wave with a 5 V amplitude, 0 V offset, and a frequency of 1 kHz.
2. Plot the input and output voltages, save an image of the plot.
3. Construct the circuit from Section 6.5.3 Item 3. Set Wavegen 1 to V_{IN1} and Wavegen 2 to V_{IN2} . Synchronize Wavegen 1 and Wavegen 2 using the dropdown in the wavegen tab of waveforms. It is located two tabs to the right of "Run all" and is defaulted to "No synchronization".
4. Plot the input (differential) and the output voltages, then save an image of the plot. Typically, scope wire 2 (with the white stripe) is connected to ground because scopes measure voltages relative to another signal. In this case, however, you connect scope wire

1 to Wavegen 1 and scope wire 2 to Wavegen 2, allowing the scope to measure the difference between them.

6.7 Write Up

Include the following in the write up.

1. Section 6.6.1
 - (a) Item 1: LTspice schematic of the circuit.
 - (b) Item 2: Plot of the input and output voltages.
 - (c) Item 3: LTspice schematic of the circuit with all calculated resistors.
 - (d) Item 4: Plot of the input and output voltages.

Discuss the two circuits constructed in lab, DC level shifter and differential to single ended converter. Explain how each circuit works at the block level and how resistor tolerance could or could not influence the output. Also explain the design choices for the circuit, Section 6.5.3 Item 3, constructed in lab. Which resistors were chosen and why?

Chapter 7

Lab 7 - Diode Applications

7.1 Objective

The objective of this lab is to introduce diodes and a few of their applications.

7.2 Materials

- Laptop with LTSpice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit

7.3 Introduction

Diodes are a fundamental circuit element whose one-directional current flow has a variety of applications. Diodes can be used as indicators (light emitting diodes (LEDs)), voltage protection elements, and rectification.

7.3.1 Diode Operation

Each diode has an anode and cathode, Figure 7.1 (a), and conducts current one way once the voltage across the diode has reached the forward voltage, often called forward bias, $V_F \approx 0.7 \text{ V}$. A diode can also conduct current in the opposite direction when the voltage across the diode is negative, often called reverse bias, where the breakdown voltage, V_{BR} , is large, usually -50 V or less. There is often a small current that flows when a diode is reverse bias, but we will neglect that effect for the purposes of this lab. The schematic symbol for an LED is similar, Figure 7.1 (b), but

indicates that light is generated when the diode is operating with a forward bias. And the I-V curve for a typical diode, Figure 7.1 (c).

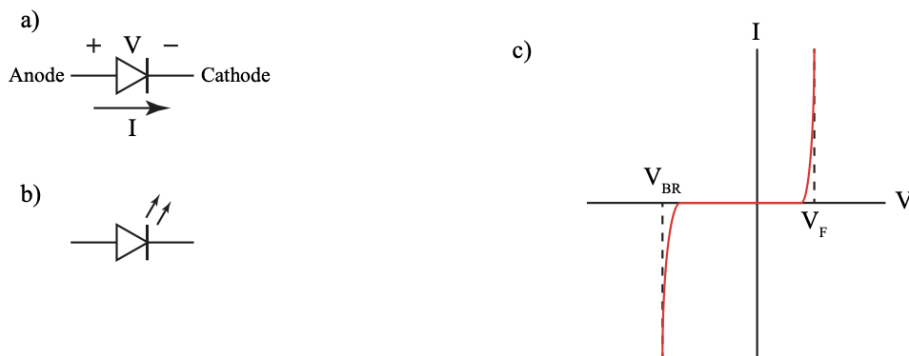


Figure 7.1: Diode schematic symbol with the anode and cathode labeled (a), schematic symbol for an LED (b), and the I-V curve for a diode (c).

7.3.2 Applications

There are a variety of applications for diodes and this lab will focus on only a few, rectifications, and clip detection.

Rectifiers

Half-wave Rectifier

Signal rectification is the process of converting an alternating signal in to its positive or negative components. Figure 7.2 is an example of a simple half-wave rectifier. After the input sine wave passes the diodes forward bias voltage, ≈ 0.7 V, the diode conducts and passes the positive half of the sine wave but doesn't pass the negative half of the sine wave because the diode is reverse bias and does not conduct.

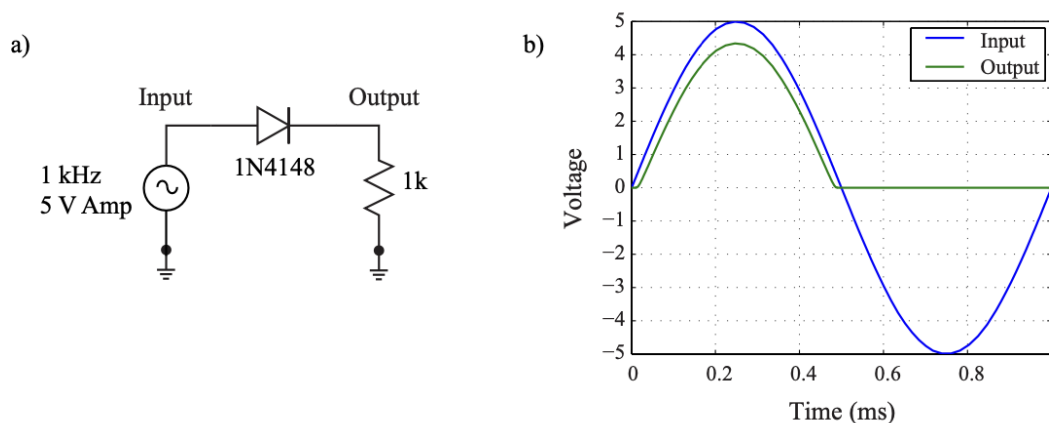


Figure 7.2: A simple half-wave rectifier (a) and the resulting input and output signals (b).

Simply adding a capacitor to the half-wave rectifier results in an AC to DC converter, Figure 7.3. The circuit initially behaves as it did previously, once the input sine wave reaches the forward bias voltage, the output follows the input. However, once the sine wave starts to decrease, the output then discharges as a function of an RC time constant, the process then repeats. For higher input frequencies and capacitances, the output can be approximated as a DC source.

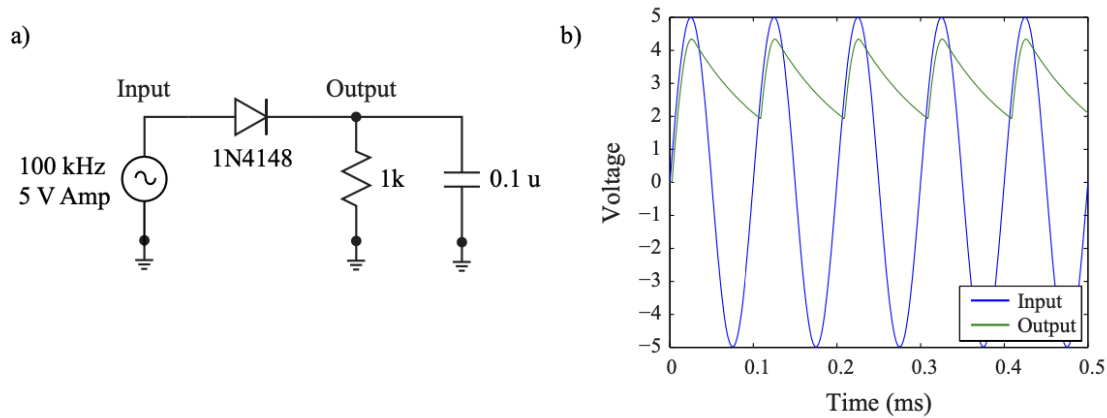


Figure 7.3: A simple half-wave rectifier but with a capacitor at the output (a) and the resulting input and output signals (b).

Full-Wave Rectifier

A full-wave (also called full-bridge) rectifier uses four diodes so that both halves of a sinusoidal input are routed to the load with the same polarity, producing a positive output for each half of the input. Because two diodes conduct in series on each half-cycle, the load voltage is reduced by about twice the diode forward drop. See Figure 7.4 for the schematic and waveforms.

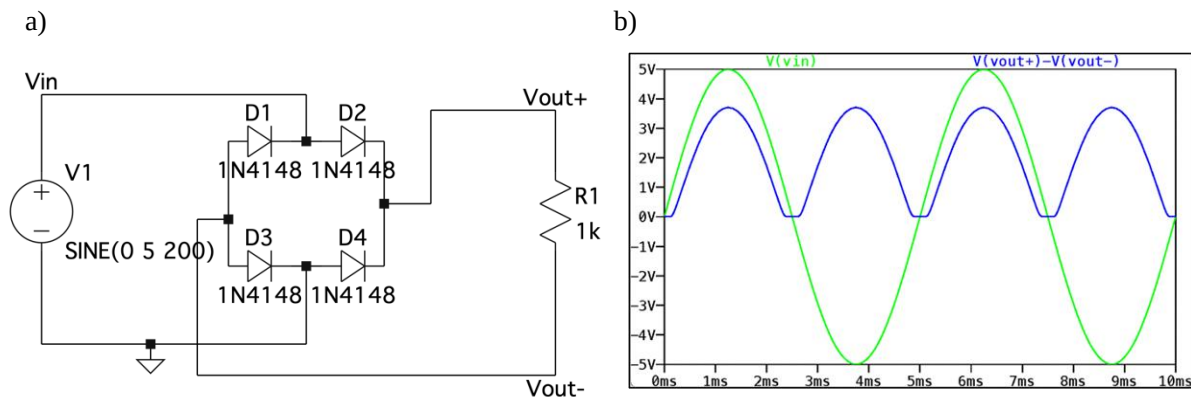


Figure 7.4: A full-bridge rectifier circuit (a) and its resulting input and output signals (b).

Clip Detection

There are often cases where the input or output of a system isn't easily accessible for test equipment. A variety of dials, indicators, and display panels are often used to report on the status of a system, examples include current draw, internal voltages, and clip detection.

Clip detection is the process of determining if a signal is clipping or not. A signal that is clipped, unless desired, can degraded the performance of the system. A clipping detection circuit is able to indicate that the output of the op amp is clipping without viewing the output on an o-scope, but instead through an LED.

Figure 7.5 is an example of a clipping detector. If the input is higher, or lower, than the reference voltage V_A , or V_B , the LED D1, or D2, illuminates indicated that the input voltage is above, or below, the reference voltage. Note that this circuit does not use op amps, but comparators. Comparators, as the name implies, compares the input at both terminals and produces a digital output, either a 1 (5 V) or 0 (0V), if the positive terminal input is larger than the negative terminal input.

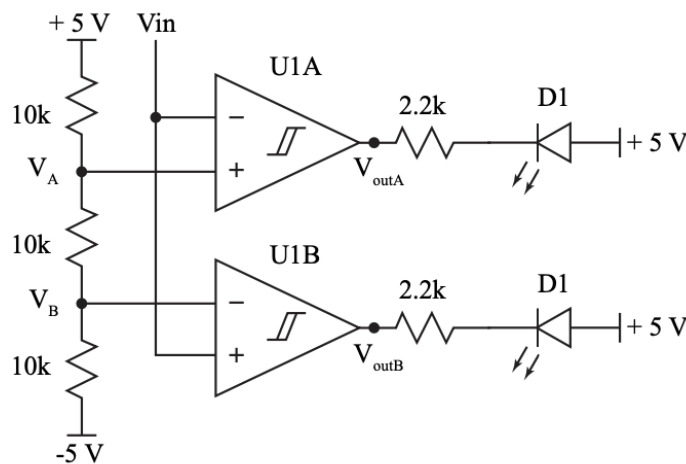


Figure 7.5: A clipping detection circuit. The input is compared to fixed voltages, V_A and V_B , by U1A and U1B respectively. If the input voltage is greater, or less, than the reference voltage, the output is pulled low and LED is illuminated.

7.3.3 Potentiometers

A potentiometer (or “pot”) is a three terminal resistor as shown in Figure 7.6. The total resistance between terminals 1 and 3 is fixed and matches the potentiometer’s rated value. Terminal 2 is

connected to a sliding contact called the wiper, which moves along the resistive path inside. By adjusting the wiper, the resistance between terminals 2 and 1 or between 2 and 3 changes.

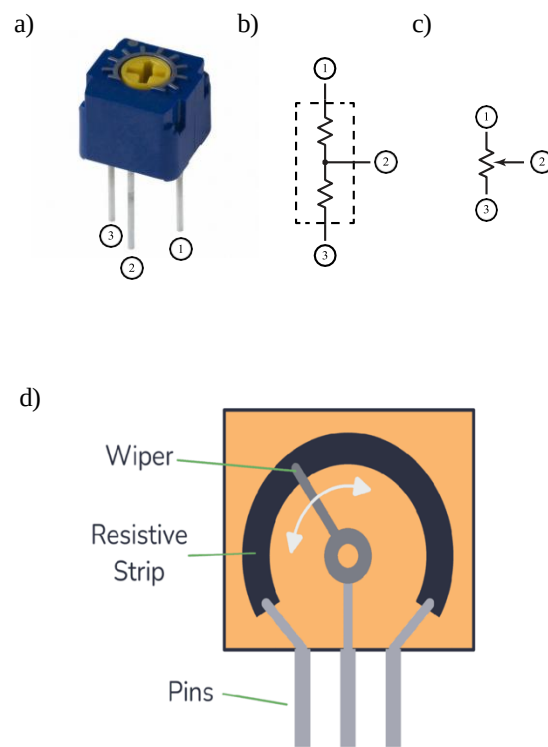


Figure 7.6: A picture of a practical potentiometer with the appropriate pins labeled (a), an equivalent circuit diagram of a potentiometer with the wiper in a fixed position (b), the common circuit schematic representation for a potentiometer (c), and a visual aid to help understand how the wiper adjusts the resistance (d).

To use a potentiometer on a breadboard, you can treat it like a standard two-terminal resistor by connecting only the center terminal (wiper) and one of the outer terminals. This allows the potentiometer to function as a variable resistor by adjusting the knob. The third unused pin can be left with no connection, do not try to bend it.

You can also use the potentiometer as a voltage divider. To do this, you would connect all three terminals: one outer terminal to Vcc, the other outer terminal to GND, and the center terminal becomes the adjustable voltage output.

In LTspice, a potentiometer is modeled using a resistor with a variable resistance. Instead of a dedicated component, you can define a resistor's value as $\{R_total * w\}$, where R_total is the max resistance (10k) and w (ranging from 0 to 1) represents the wiper position. For example, if you want the pot to be 5k Ω , the value of w would be 0.5 and the spice directive would be:

```
.param w=0.5
```

You can also calculate the resistance you need and just assign that value to the resistor in your schematic. To recall how variables work in LTspice, look back to the Monte Carlo simulations in Lab 2 where we define resistor tolerances.

7.4 Big Picture

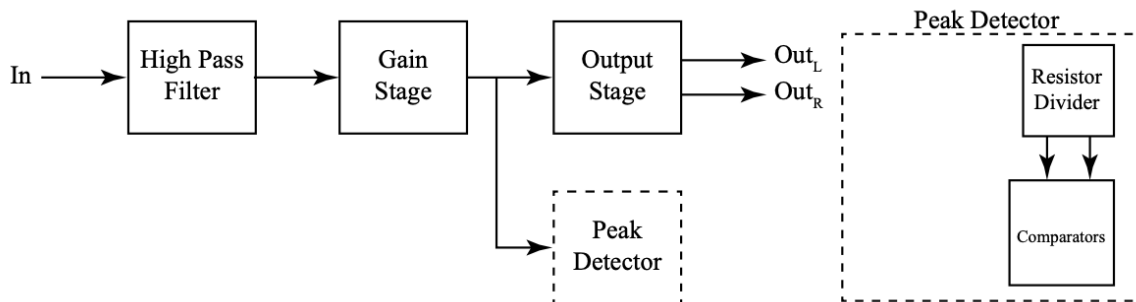


Figure 7.7: Big picture with emphasis on the peak detector.

This lab focuses on diode circuits which will help to form the comparator circuits used in the peak detector for the final project.

7.5 Pre-Lab Requirements

Complete the following before coming to lab.

7.5.1 LTspice Simulations

Using LTspice, complete the following.

1. Build the circuit in Figure 7.2 (a). Set the input to a 1 kHz, 5 V amplitude sine wave and run a transient simulation with a stop time of 1m (.tran 1m) and plot the input and output. To use the 1N4148 diode model, right click on the diode after placing it, Pick New Diode, and then choose the 1N4148 model (Mfg. OnSemi). Save an image of the circuit and the plot of the input and output voltage.
2. Build the circuit in , Figure 7.3 (a). Set the input to a 100 kHz, 5 V amplitude sine wave and run a transient simulation with a stop time of 50u (.tran 50u) and step through possible capacitor values using the following spice directive:

```
.step param C list 0.001u 0.01u 0.1u 1u
```

Save an image of the circuit and the plot of the input and output voltage.

- Rebuild the circuit in Figure 7.4 (a) in LTspice and plot the input and output voltages. Use the 1N4148 diode and a sinusoidal input voltage with 0 V offset, 5 V amplitude, and 200 Hz frequency. When plotting the output voltage, you must plot the difference between the two nodes of the load, just like in Figure 7.4 (b). Use .tran 10m for simulation. Take screenshots of the schematic and voltage waveforms.
- Build the circuit in Figure 7.5. Use the default LED model in LTspice, and download "slcj016.zip" file from Lab 7 folder for the LM393 comparator model. Ensure the unzipped file ends correctly by opening it in a text editor. The final line should be ".ENDS" with no extra characters following it. If any exist, remove them, save the file, then proceed to import to LTspice.

Import the model into LTspice using the "LTspice Component Import" documents in the lab folder. You can also find video instructions [here](#). You don't need to adjust the component shape; it could lead to potential errors.

Power the LM393 with ± 5 V. Set the input voltage to a 1 kHz, 5 V amplitude sine wave and run a transient solution with a stop time of 2m (.tran 2m). Plot the input voltage and the current through each diode. Save an image of the circuit and the plot.

- Build the circuit in Figure 7.8. Power the LM393 and TLV272 with ± 5 V. Set the input voltage to a 1kHz, 0.1 V amplitude sine wave and run a transient solution with a stop time of 2m. Choose the value for R1, 10k pot, (vary the 'w' parameter or just hard code the value) so that the diodes conduct current and illuminate. Plot the input voltage, output of the op amp, positive and negative inputs to the comparator, and the current through both LEDs. (6 items total). Save an image of the circuit and the plots.

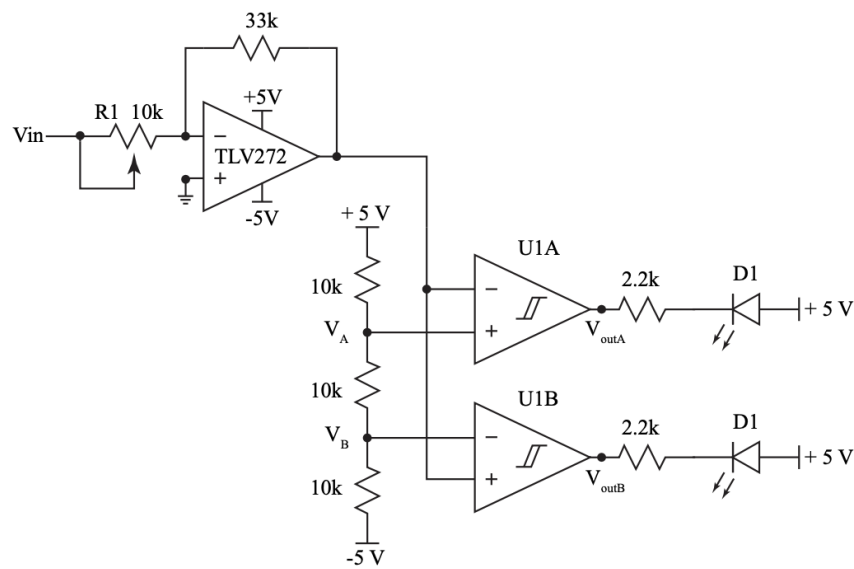


Figure 7.8: A variable gain amplifier, R1 is a potentiometer that can be varied, where the output is checked for clipping.

7.5.2 Breadboard Implementation

1. Read the [datasheet](#) for the LM393, the pinout for the chip is on page 3.
2. Build the circuit in Figure 7.5 on a breadboard. It is important to note that the pinout for the comparator chip is the same as the op amp. It works differently but should be connected the same as the op amp. This means you should still connect pin 8 to +5V and pin 4 to -5V (not GND) just as you have done with the op amp.

Set the input to a 0.5 Hz, 5 V amplitude sine wave. Use the two red LEDs in your lab kit, they're bagged separately. The anode for the LEDs is the longer of the two leads. The diodes will illuminate whenever the input voltage is above or below the reference voltage and should appear to blink in turn. Save the circuit to show your lab instructor at the start of lab.

7.5.3 Pre-Lab Checklist

The following must be completed at the start of lab.

1. Section 8.5.1
 - (a) Item 1: Half-wave rectifier, image of the circuit and the plot of the input and output voltage.
 - (b) Item 2: Half-wave rectifier with output smoothing, image of the circuit and the plot of the input and output voltage.
 - (c) Item 3: Full-wave rectifier, image of the circuit and plot of the input and output voltage.
 - (d) Item 4: Clipping circuit, image of the circuit and the plot.
 - (e) Item 5: Variable gain amplifier with clip detector, image of the circuit and the plot.
2. Section 8.5.2
 - (a) Item 2: Operational clipping circuit demo.

7.5.4 Breadboard Implementation

1. Plot the input voltage and the voltage at the output of either comparator, Figure 7.5, on the o-scope and save an image of the display.

2. Build the circuit from Section 8.5.1 - Item 1. Plot the input and output on the o-scope and save an image of the display.
3. Build two versions of the circuit from Subsection 8.5.1 - Item 2, one with where the capacitance is 0.1 μF and the other is 0.01 μF . Plot the outputs of both circuits on the o-scope and save an image of the display.
4. Implement the full-bridge circuit from Subsection 8.5.1 - Item 3. Set Wavegen 1 to a 200 Hz, 5 V amplitude sine wave. When measuring the output, connect Scope 2+ to Vout+ and Scope 2- to Vout- to get the differential signal. Plot the outputs of both circuits on the o-scope and save an image of the display.
5. Build the circuit from Subsection 8.5.1 - Item 5. Set the input to a 0.5 Hz, 0.1 V amplitude sine wave. Vary the potentiometer, increase the gain, so that the LEDs illuminate when the op amp output is greater than the reference voltages. Plot the output of the gain amplifier and the output of either comparator on the o-scope and save an image of the display.

7.6 Write Up

Discuss the functionality and operation of each circuit built in lab, include your captured waveforms and corresponding LTspice schematics. By this point, you have worked with both op amps and comparators. Compare their similarities and differences. When would you choose an op amp over a comparator, and vice versa, based on application?

Refer back to the clipping detection circuit in Figure 7.5. For the final project, you will be asked to design a circuit where one LED turns on when the input voltage exceeds 1 V and the other LED turns on when the input voltage exceeds 4 V to indicate clipping. Calculate the resistor values needed to set the correct reference voltages using either $\pm 5\text{ V}$ supply or +5 V and GND supply for the three-resistor ladder. You do not need to get exactly 1 V and 4 V, get as close as you can only using 3 resistors. Explain your reasoning.

Chapter 8

Lab 8 – Filters

8.1 Objective

The objective of this lab is to introduce the concept of filters and explore their applications through examples of passive, active, and cascaded filters.

8.2 Materials

- Laptop with LTSpice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit

8.3 Introduction

Time Domain vs Frequency Domain

Up to this point, the labs have focused on signals in the time domain. In the time domain, we observe how a signal varies over time using an oscilloscope. For example, if a signal has a specific frequency, you see it as a repeating waveform on the screen, with the timing set appropriately.

When we move into the topic of filters, we start looking at signals in the frequency domain. In this domain, we no longer ask “what does the signal look like over time?” but instead, “which frequencies does the signal contain, and how does the circuit affect them?” In other words, we care about how a system responds to different frequencies, not how the signal looks versus time.

A helpful analogy is to think of a filter like a colored lens:

- In the time domain, you might describe how bright or dim a light is changing over time.
- In the frequency domain, you think about which colors (frequencies) the lens lets through, and which does it block.

Filter Types

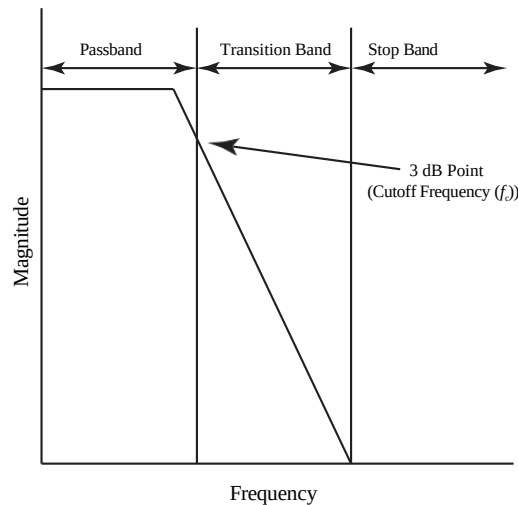


Figure 8.1: Example of response for a generic lowpass filter. The filter passes low frequencies, thus the name, and attenuates higher frequencies.

Figure 8.1 shows a typical lowpass filter response in the frequency domain. As the name suggests, a lowpass filter allows low-frequency signals to pass through, while higher frequency signals are gradually attenuated as they enter the transition and stop bands.

A highpass filter works the opposite way, it attenuates low-frequency signals while allowing high-frequency signals to pass through with little to no attenuation.

Other common filter types include band-pass filters, which only pass signals within a specific frequency range and attenuate frequencies outside that range, and notch (band-reject) filters, which block a specific range of frequencies while allowing frequencies above and below that range to pass. Lastly, an all-pass filter passes all frequencies but alters the phase of the signal without affecting its amplitude. For more information on these filters, you are encouraged to read about them [here](#) in Section 8.2.

Cutoff Frequency

The point where the filter begins to noticeably attenuate the signal is called the cutoff frequency or 3 dB point, the frequency where the output drops to 70.7% (or -3 dB) of its maximum value. This cutoff frequency is chosen based on the design requirements to control how much attenuation occurs at a specific frequency.

It is important to note that the cutoff frequency is not always located at exactly -3 dB on the plot. The -3 dB point is a relative measurement, defined as the frequency where the gain drops by 3 dB below the maximum gain of the filter. For instance, if an active filter has a maximum gain of 10 dB, the cutoff frequency will be found where the gain falls to 7 dB, not at -3 dB. This relative rule applies regardless of whether the filter is passive or active.

8.3.1 Passive First Order RC Filters

Figure 8.2 shows two simple resistor-capacitor (RC) filter configurations and their resulting frequency responses. These are passive filters, meaning they do not contain active components like transistors or op amps and do not require an external power supply to function. Additionally, they are first order filters since each circuit contains only one RC pair.

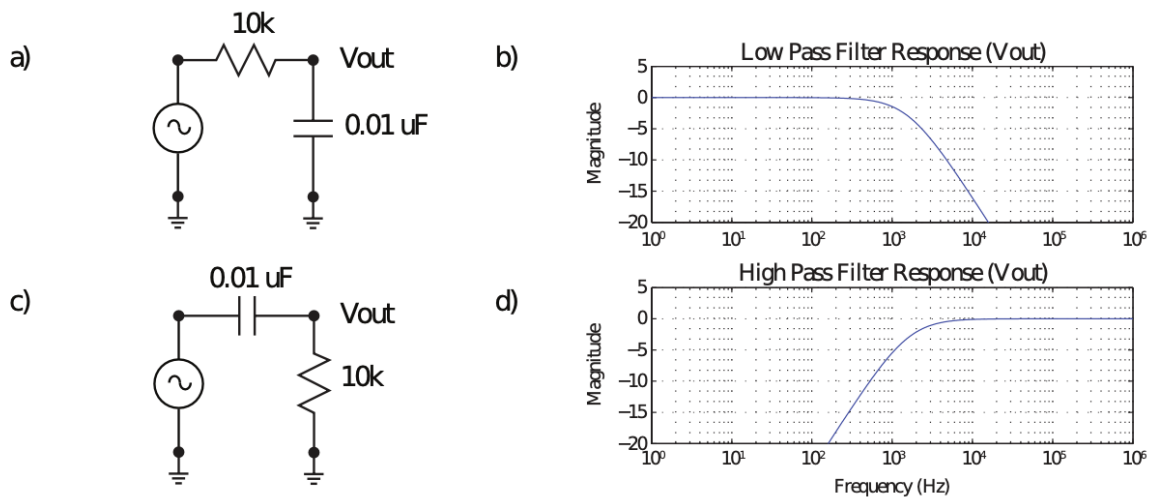


Figure 8.2: An example of a simple RC low pass filter (a) and the resulting frequency response (b). Also, a simple RC highpass filter (c) and the resulting frequency response (d).

The behavior of these circuits becomes clearer when considering how capacitors respond at different frequencies:

- At low frequencies, a capacitor behaves like an open circuit because its impedance $Z_c = \frac{1}{j\omega C}$ is large when $\omega = 2\pi f$ is small.
- At high frequencies, a capacitor acts like a short circuit because Z_c decreases as ω increases.

With this in mind, it should make sense why lowpass filters allow lower frequencies to pass. As the frequency increases, the capacitor creates a stronger connection between ground and Vout. This prevents less of the input signal to reach the output, thus attenuating it. The same principles can be applied to the highpass filter.

Rather than working through the full transfer function for filter design and analysis, we will focus on key characteristics, such as the cutoff frequency. This is the frequency at which the filter begins to significantly attenuate signals. For a simple RC filter, the cutoff frequency is determined by:

$$f_c = \frac{1}{2\pi RC} \quad (8.1)$$

For the examples in Figure 8.2 (a) and (c), the 3 dB point is calculated to be:

$$\frac{1}{2\pi(10k)(0.01\mu)} = 1.5916 \text{ kHz}$$

Which matches the frequency response for both configurations.

8.3.2 Active RC Filters

There are a variety of active filter topologies, but a simple implementation is to use an appropriately placed capacitor in conjunction with a typical inverting amplifier configuration, Figure 8.3 (a) and (b).

Aside from requiring an external power supply, active filters differ from passive filters in their amplitude response. In passive filters, the output amplitude can never exceed the input amplitude, meaning the maximum gain is 1 or less. In contrast, active filters use an op amp to provide amplification, allowing the output to be greater than the input amplitude. This means active filters not only shape the frequency response but can also amplify the signal at the same time.

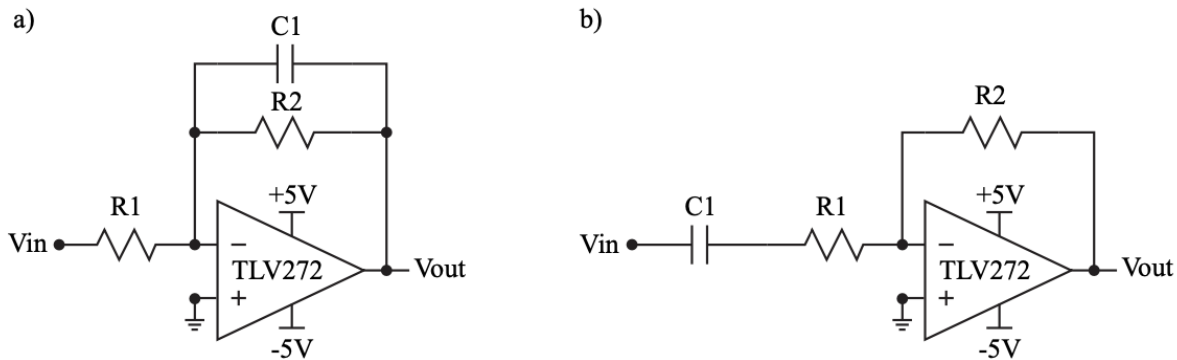


Figure 8.3: An example of an active RC low pass filter (a) and an active RC high pass filter (b).

The design of either the active RC low pass or high pass is also fairly simple for the first order case. A low pass filter has the following equations:

$$\frac{V_{OUT}}{V_{IN}} = -\frac{R_2}{R_1} \left(\frac{1}{j\omega C_1 R_2 + 1} \right)$$

$$H_0 = -\frac{R_2}{R_1}$$

$$f_0 = \frac{1}{2\pi C_1 R_2}$$

where H_0 is the passband gain and f_0 is the 3dB frequency. Similarly, the high pass filter has the following equations:

$$\frac{V_{OUT}}{V_{IN}} = -\frac{R_2}{R_1} \left(\frac{j\omega C_1 R_1}{j\omega C_1 R_1 + 1} \right)$$

$$H_0 = -\frac{R_2}{R_1}$$

$$f_0 = \frac{1}{2\pi C_1 R_1}$$

8.3.3 Cascaded Filters

While single-stage (first order) filters are useful, they often do not attenuate unwanted frequencies fast enough for many practical applications. To achieve a sharper transition between the passband and stopband, or to build more complex frequency responses, multiple filter stages can be connected in series—a technique known as **cascading**.

When filters are cascaded, the output of the first stage serves directly as the input to the second stage. The overall transfer function of N non-interacting cascaded stages is the product of their individual transfer functions:

$$H_{\text{total}}(\omega) = H_1(\omega) \cdot H_2(\omega) \cdot \dots \cdot H_N(\omega)$$

Connecting two passive RC filters directly in series creates a **loading effect**. The input impedance of the second stage draws current from the first stage, altering its effective RC time constant. As a result, the cutoff frequency shifts away from its calculated theoretical value. Active filters built with op amps overcome loading issues entirely. Because an op amp has very high input impedance and extremely low output impedance, it acts as a buffer between stages. This prevents stage-to-stage loading, ensuring each filter stage operates independently and predictably. Below is an example of cascaded high-pass filters using the passive RC topology shown above.

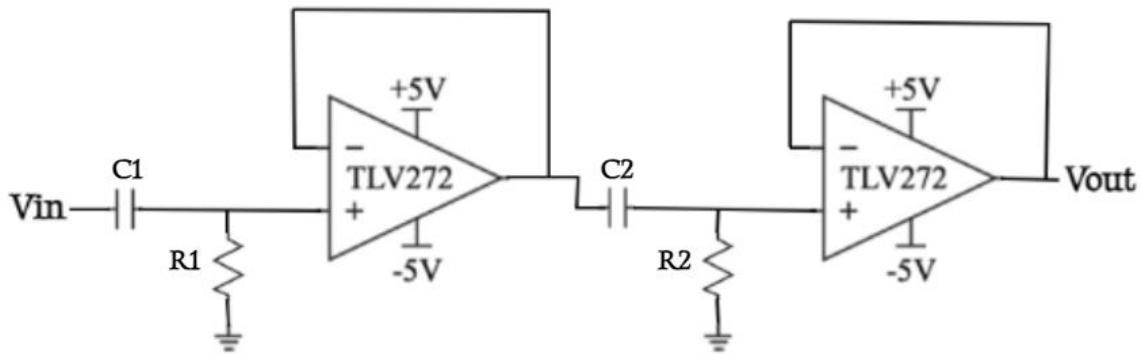


Figure 8.4: An example topology of a cascaded RC high-pass filter.

8.4 Big Picture

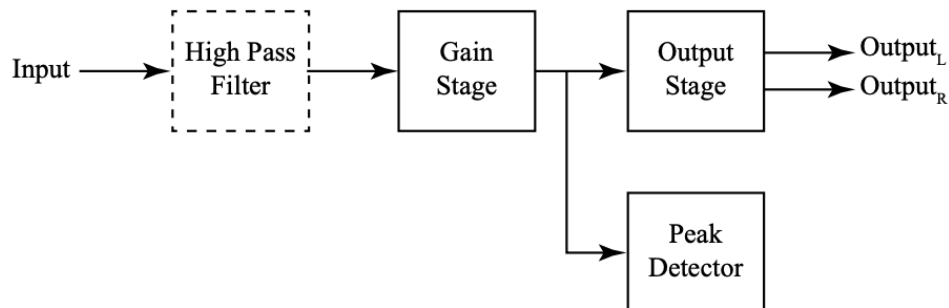


Figure 8.5: Big picture with emphasis on the high pass filter.

This lab focuses on filter circuits which will help to create the high pass filter used in the final project used to remove the large DC offset in the source signal.

8.5 Pre-Lab Requirements

Complete the following before coming to lab.

8.5.1 LTspice Simulations

Complete the following using the TLV271 spice model, always power the op amp with $\pm 5V$ for V_{dd+} and V_{dd-} .

1. Review the LTspice tutorial for [AC Analysis](#). Also please note that the component values in the example image in 8.3.1 are different than the values needed for the following instructions. For your reference: $0.001\ \mu\text{F} = 102\ \text{J}$, $0.01\ \mu\text{F} = 103\ \text{J}$, $0.1\ \mu\text{F} = 104\ \text{J}$, $0.033\ \mu\text{F} = 333\ \text{J}$, $0.047\ \mu\text{F} = 473\ \text{J}$.
2. Build a simple lowpass filter, Figure 8.2 (a), but set $R = 10\ \text{k}\Omega$ and $C = 0.001\ \mu\text{F}$. Set the voltage source to an AC amplitude of 1 and run an AC analysis with the following settings: Decade, 100, 1, 1Meg. Save an image of the circuit, a plot of the output, and table the 3 dB frequency.
3. Build an active RC highpass filter, Figure 8.3 (b). Set $C_1 = 0.1\ \mu\text{F}$ and determine the resistor values for a gain of 10 (20 dB) and a $f_0 = 482.3\ \text{Hz}$ using only values in your lab kit. Set the voltage source to an AC amplitude of 1 and run an AC analysis with the following settings: Decade, 100, 1, 1Meg. Table the 3 dB frequency. Save an image of the circuit, a plot of the output, and table the 3 dB frequency (3 dB from the passband) for submission to Canvas.
4. Build cascaded RC highpass filters as shown in Figure 8.4. Choose resistor and capacitor values from your parts kit to attenuate 240Hz noise. Note you should have the same resistor and capacitor values in both stages of the filter, meaning R_1 should be the same as R_2 and C_1 should be the same as C_2 . In the first stage of the filter 240Hz should be attenuated by at least -3dB, and in the final output, the attenuation should be double of the first stage. Table your resistor and capacitor values as well as the expected attenuation at 240Hz in each stage. Set the voltage source to an AC amplitude of 1 and run an AC analysis with the following settings: Decade, 100, 1, 100k. When plotting the bode plots, display the output of the first stage and the second stage on the same plot. Add a cursor to each plot trace to clearly show the attenuation at 240Hz. Save an image of the circuit and a plot of the output. (You can use the following links to determine what resistors and capacitors you should choose: <https://www.calculator.net/resistor-calculator.html>, <https://www.digikey.com/en/resources/conversion-calculators/conversion-calculator-low-pass-and-high-pass-filter>)
5. Use the schematic below (Figure 8.6) to build a RC band reject filter. Once again, choose resistor and capacitor values from your parts kit to attenuate 240Hz by at least -3dB. Use $8.2\ \text{k}\Omega$ resistors for R_3 , R_4 , and R_5 . Table your chosen resistor and capacitor values as well as the expected cutoff for the low-pass and the high-pass stages. Set the voltage source to an AC amplitude of 1 and run an AC analysis with the following settings: Decade, 100, 1, 100k. Add a cursor to the trace to clearly show the attenuation at 240Hz. Save an image of the circuit and a plot of the output. (Hint: For the high pass component, choose a cutoff frequency at least 80 Hz above the target frequency, and for the low pass component choose a cutoff frequency at least 80 Hz below the target frequency). **Note: Your bode plot might not be the most attenuated at 240Hz. That is okay! Your only goal is to ensure 240Hz is attenuated by at least -3dB while frequencies significantly lower or higher than**

240Hz are in the passband.

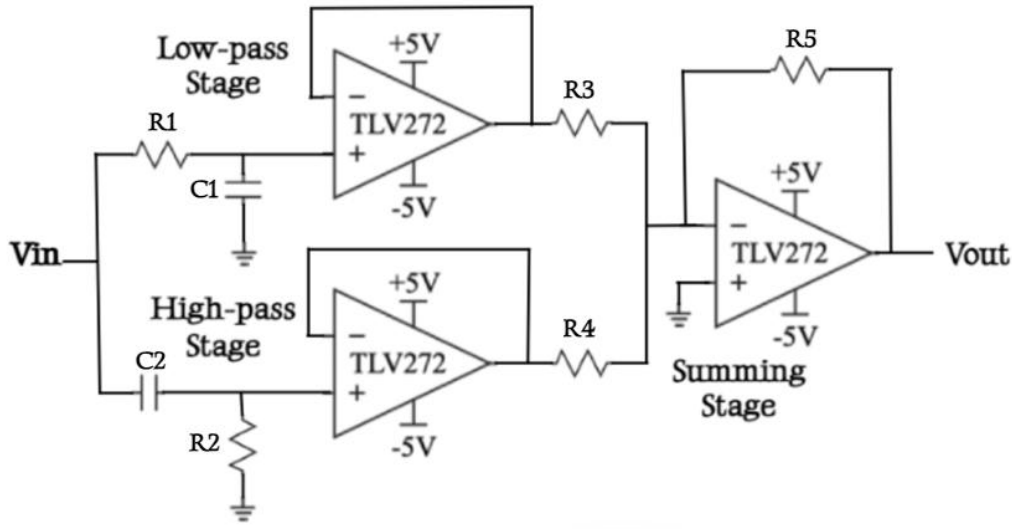


Figure 8.6. Topology for a band reject filter.

8.5.2 Breadboard Implementation

1. Review the Digilent tutorial for the [Network Analyzer tool](#).
2. Build the active RC high pass filter from Section 8.5.1 - Item 3 on a breadboard. Use a TLV272 op amp and power it with $\pm 5V$ rails and wire up the Analog Discovery to use the network analyzer tool. Connect CH1+ and W1 to the input and CH2+ to the output.
3. Set the start frequency to 100 Hz, stop frequency to 100 kHz, and the steps or samples to about 100. Set the amplitude to 100mV from 1V and the top to 30dB for this case. Click run once to generate the filter's frequency response. Save the circuit to show your lab instructor at the start of lab.

8.5.3 Pre-Lab Checklist

The following must be completed at the start of lab.

1. Section 8.5.1
 - (a) Table of 3 dB frequencies (two values).
 - (b) Item 2: Simple RC lowpass filter - image of the circuit and a plot of the output.
 - (c) Item 3: Active RC highpass filter - image of the circuit and a plot of the output.

- (d) Item 4: Cascaded RC highpass filter - image of the circuit and a plot of the output.
- (e) Item 5: RC band reject filter - image of the circuit and a plot of the output.
- (f) Table of resistor values, capacitor values, and expected cutoff frequencies for the cascaded filter and the band reject filter.

2. Section 8.5.2

- (a) Item 3: Active RC highpass filter - image of the circuit and a plot of the output.

8.6 In-Lab Requirements

For all active circuits, use a TLV272 op amp and power it with ± 5 V rails. Remember to use the RC values as indicated in the instructions, not the images.

1. Set the number of samples to 1000 and re-run the network analyzer for the demo active highpass filter. Save an image of the output and measure the 3 dB frequency.
2. Build the simple lowpass filter from Section 8.5.1 - Item 2 ($10\text{k}\Omega$ resistor and $0.001\mu\text{F}$ capacitor). Run the network analyzer with an amplitude of 1V and top set to 10dB, measure the 3 dB frequency, and save an image of the output.
3. Build the cascaded highpass filter from Section 8.5.1 – Item 4. Use the resistor and capacitor values from the prelab. If you chose a higher cutoff frequency, the noise will be more attenuated, but so will the “bassy” components of the audio file.
4. First, use the output of the first stage in the filter as the Vout for the network analyzer and run a network analyzer from 10 Hz to 100 kHz with 150 steps first to verify your plot and then with 1000 steps while leaving the rest of the settings as default (amplitude of 1V and offset of 0V). Take a screenshot of the Network analyzer showing the 240Hz frequency using a vertical measurement line. Table this value.
5. Now use the output of the entire cascaded filter as the Vout for the network analyzer. Take a screenshot of the Network analyzer from 10 Hz to 100 kHz with 150 steps first to verify your plot and then with 1000 steps while leaving the rest of the settings as default (amplitude of 1V and offset of 0V). Make sure to showcase the steeper roll-off of the cascaded filter and add a vertical measurement line to show the attenuation at 240Hz. Once again table this value.
6. Connect the Vout to the audio jack from the parts kit. When connecting the audio jack, make sure the Vout signal is connected to the left and right pins, while the middle pin is connected to ground. Download “clean.wav” (from Canvas Lab manual’s Lab 8 folder) and play it to a speaker with the audio jack without the cascaded filters. To play a “.wav” file with Waveforms,

use the Wavegen and change the Enable setting to “Play” from “Simple”. Then click on import and click on the desired file while leaving all other settings as default.

7. Once you hear “clean.wav”, change the imported audio file to “noisy.wav” and once again play the audio file without running the signal through the cascaded filter (connect the input directly to the audio jack). Now you should know what the normal audio file and the noisy audio file sound like.
8. Run the “noisy.wav” audio file through the cascaded filter (connect the output to the audio jack) and listen if you can hear a difference in the noise level!
9. Build the band reject filter from Section 8.5.1 – Item 5. Use the resistor and capacitor values from the prelab. Take a screenshot of the Network analyzer tool with a vertical measurement line at 240Hz that shows at least -3 dB attenuation and also table this value. Change the Network analyzer to start at 10 Hz to 100 kHz with 150 steps first to verify your plot and then with 1000 steps while leaving the rest of the settings as default (amplitude of 1V and offset of 0V). Also, change the magnitude bottom to -10 dB while leaving the top at 10 dB. **Note: Your bode plot might not be the most attenuated at 240Hz. That is okay! Your only goal is to ensure 240Hz is attenuated by at least -3dB while frequencies significantly lower or higher than 240Hz are in the passband.**

8.7 Write Up

Include the following in the write up.

- 1 Table of experimental 3 dB frequencies and their associated percent errors when compared to the pre-lab.
- Network analyzer images for each circuit including vertical measurement lines at 240Hz for the cascaded filter and the band reject filter.
- 1 Table of attenuation at 240Hz for the cascaded filter and the band reject filter.

Suppose you're working with a system to measure a small bio-signal (such as an ECG or EMG) where accurate observation of the signal is critical. However, you notice that your measurements are contaminated with 60 Hz power line noise, making it difficult to clearly see the bio-signal.

- What type of filter would you design to suppress the 60 Hz interference while still preserving the desired signal? What cutoff frequency would you choose?
- Would you choose a passive or active filter, and why?

Filters can be treated like circuit building blocks and, much like resistors or capacitors, can be placed in series or parallel to create more complex behavior.

- What are the potential effects of combining filters in this way?
- What happens when a signal passes through a lowpass filter followed by a highpass filter? How does this combination affect the overall frequency response?

Think about how the individual cutoff frequencies of each filter might interact, and what type of overall filter this could resemble depending on how the filters are configured.

Chapter 9

Lab 9 - Final Project

9.1 Objective

The objective of the final project is to bring together concepts taught in previous labs into a single circuit.

9.2 Materials

- Laptop with LTspice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit with TLV272 and LM393
- Speakers with 3.5mm connector

9.3 Introduction

The nature of this lab will be different to accommodate the final project. With a total score that accounts for 20% of the entire lab grade, the final project is required to pass the course. There is no quiz for the final project and the breakdown is as follows:

- 20% Pre-lab
- 40% In-lab Demonstration (This includes building your final circuit on a provided printed circuit board)

- 40% Write up

The pre-lab requires a functional LTspice schematic with the correct outputs, which is easier than it may sound. Source files for the simulation (Jingle4.wav and Jingle19.wav) are available on Canvas under the Lab 9 folder. To import the WAV files into LTspice, follow the instructions [here](#).

The final simulation, must use the Jingle19.wav file and a 19-second transient simulation (.tran 19). Since the simulation takes several minutes to complete, a shorter file, Jingle4.wav (4 seconds), is provided for faster testing.

Failure to complete the pre-lab will result in a zero for 20% of the final project and exclusion from the first lab sessions. However, you will not receive a zero for the entire project.

Lab sessions for the final project span two weeks and are dedicated to demonstrating your working breadboard circuit with in-lab speakers. As there aren't enough speakers for students to take home, they must remain in the lab. The circuit must be checked off by the end of the second lab session.

Additional details about the project write-up are provided below.

9.3.1 Audio Amplifier

The final project is a basic audio amplifier with a block diagram shown below in Figure 9.1. A simple jingle with a small amplitude, mV, and a large DC offset, 0.5 V, will serve as the input. The system functions as follows:

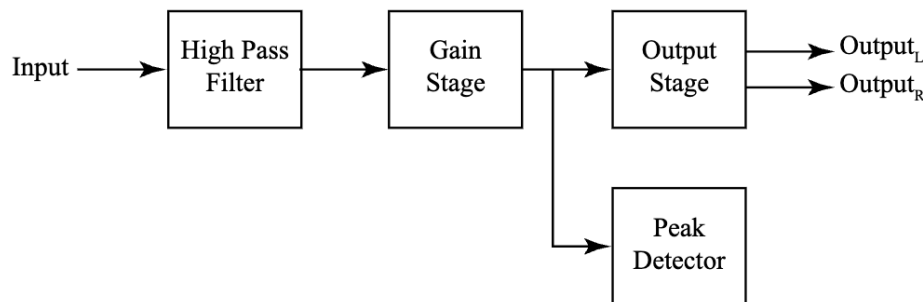


Figure 9.1: Audio amplifier block diagram.

1. The high pass filter removes the DC offset from the jingle, effectively AC coupling the signal into the system.
2. A gain stage is used to amplify the jingle to a range of volts from mV so that it can be heard from the speakers.

3. An output stage must buffer the output to the speakers, the outputs do not need to be offset in phase.
4. A peak detector is also required to indicate when the output is in the right range, defined by two thresholds. When the output passes the first threshold (LED 1 turns on), the output is operating in the normal linear range with an output on the order of Volts, and the second threshold (LED 2 turns on), where the output is about to start clipping. See the different LED states in Table 9.1. Note that this is a slight variation on the circuit used in Lab 7.

State	LED 1	LED 2
Input < Threshold 1 and 2	Off	Off
Threshold 1 < Input < Threshold 2	On	Off
Threshold 1 and 2 < Input	On	On

Table 9.1: Different states for the peak detector

9.3.2 Recommendations

A few words on some of the individual sections.

Design Considerations

- You are restricted to the components in your lab kit.
- High Pass Filter:
 - Choose the 3 dB frequency carefully. The audio band is 20 Hz to 20 kHz, so a cutoff frequency in the kHz range will attenuate important parts of the jingle. Select a cutoff frequency between 80 Hz and 1kHz for best results.
- Gain Stage:
 - The require gain should be high enough to go from mV to V. This should occur in two stages:
 1. The High Pass Filter with a fixed gain.
 2. A variable gain amplifier using a potentiometer as the input resistor to allow for an adjustable gain. Review Lab 7 to see how to implement the potentiometer in LTspice.

- Keep in mind that when frequencies responses are in series, they multiply rather than add. The total minimum gain should be lower than 100 to avoid issues with reaching the Cutoff state.
- Peak Detector:
 - Your thresholds should be 1 V and 4 V. This circuit should have been made in the Write Up for Lab 7. The thresholds do not need to be exactly 1 V and 4 V, get as close as you can using 3 resistors for the resistor ladder. The PCB only has space for 3 resistors.
 - There are three states:
 1. Cutoff: Both LED's are off when the input is < 1 V.
 2. Linear: One LED is on when the input is between 1 V and 4 V. This indicates normal operation and gives the best sound quality.
 3. Saturation: Both LED's are on when the input is > 4 V. This produces a highly distorted audio.

Simulation and Debugging

- LTspice Simulation
 - Before breadboard implementation, ensure that your simulation works correctly.
 - The simulation with the 19-second jingle file will take several minutes to run. Start with a simpler setup:
 - Use a 10 mV amplitude, 1kHz sine wave with a 0.5 V DC offset.
 - Perform a transient simulation (.tran 1m) to test.
 - Once you confirm that the DC offset is removed and gain is added correctly, proceed with the 4-second jingle file. After that is also confirmed, proceed with the 19-second jingle to produce the final result.
 - Be organized!
 - Use net nodes instead of separate power supplies for your active components.
 - Keep the circuit layout linear and avoid creating a maze of wires.

Project Timeline

- The project should be completed in two sessions.
 - Session 1: Submit pre-lab and complete first breadboard demo.
 - Once you've demoed, capture screenshots of the necessary waveforms for the write up **before** disassembling.
 - Week 2: Solder your circuit onto a provided PCB and demo the completed project.
 - The Write Up report will be due towards the end of the semester. See the canvas assignment for the exact date.

9.4 Pre-Lab Requirements

The following must be completed and submitted by the start of lab.

1. Image of the spice schematic, a plot of the input, and all subsequent outputs (outputs of each individual stage) for a transient simulation for Jingle19. Include the plot of the input/outputs and a circuit schematic, nothing else. The schematic and plot should be appropriately labeled.

Images that are unclear or vague will receive little to no points. Your pre-lab must include the following in different plot planes:

- Input voltage
- Output voltage of the high pass filter
- Output voltage of the variable gain amplifier
- Output voltage of the buffer stage (at least one),
- Output current of both LEDs.

Set the gain so that the circuit operates in the linear state with one LED on. Not all of the output must be above 1 V, get as much as you can without breaking the 4 V threshold.

9.5 In-Lab Requirements

Unlike previous labs, the final project lab spans two weeks (two lab sessions). The final project in-lab portion is only complete once a student has demoed a working circuit on both the breadboard and on a fully soldered PCB. There are two steps to demonstrating a working circuit:

1. Using an input of a 10 mV amplitude, 1 kHz sine wave with a 0.5 V DC offset, demonstrate that the output can be varied and the peak detectors functions properly. This means that all three operation states can be shown (cutoff, linear, and saturation).
2. Using the 19 second jingle file as an input, demonstrate that the output can be heard on both speakers and the peak detector functions properly.
 - To import the jingle to Waveforms, go to Wavegen and change the wave type from Simple to Play. Then import the jingle19.wav file and don't change any of the settings. Hit run to start playing the jingle.
 - The 3.5 mm connector in your kit is used to connect to the speakers. It has 3 pins: the middle pin is for ground, and the outer pins are for the outputs of your buffers. There is no distinction between left and right pins since the output signal is the same for both channels.

9.6 Write Up

The write up is also different for the lab project, it takes the form of a formal report with the following section requirements:

Abstract

Six to ten sentences (more than ten or less than six will get points off). Give a brief overview of the circuit built for the project and explain how it works. Weak abstracts will not be graded.

Procedure

Four to six sentences per stage. Discuss the design of each individual stage. Include a block diagram, do not copy the block diagram from the lab manual.

Results

Include all your results here. All figures and tables must include a number and a caption, figures without numbers and captions will not be graded. Screenshots that are poorly taken will not be graded.

When taking your screenshots, ensure that the plot settings on the side are not visible. Look at Figure 1 (a) as an example. Figure 1 (b) is an example of what not to include.

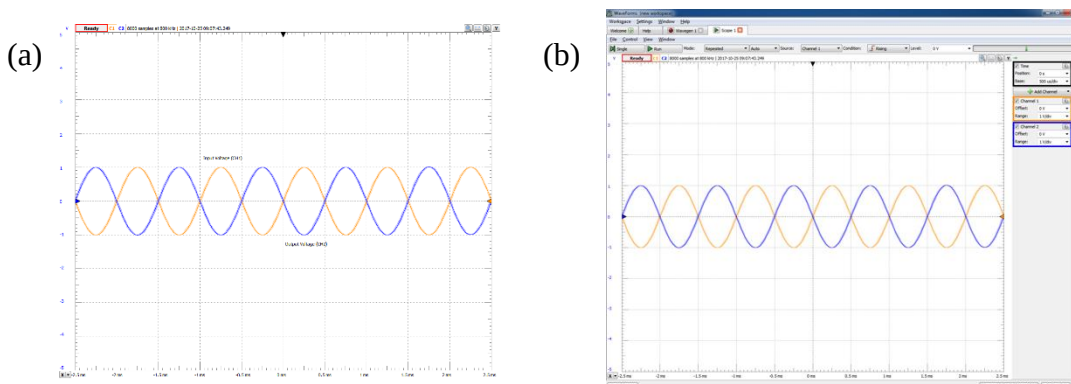


Figure 1: Example plots with two signals out of phase. Note the settings being excluded in (a) but not in (b).

1. Include your LTspice schematic of the final circuit. All nodes should be labeled appropriately, for example, Gain Stage output, and the diagram should be clearly visible so that node names and component values are readable. Schematics that aren't clearly visible or appropriately labeled will not be graded.
2. Create a parts list. Table all the components used and include their quantities, values, prices, and total price. Neglect items such as wires, jumpers, the Analog Discovery, 3.5mm connector, and speakers.
3. All waveform images (items 4-6) must be taken when the circuit is in linear operation (one LED on). To ensure the gain is set correctly, use the demo input (10 mV, 1 kHz sine wave with 0.5 DC offset) and change the potentiometer such that the final output voltage is around 2 V. Once this potentiometer position is set, proceed with the following instructions.
4. Plot of the input (CH1+) and (CH2+) output (either buffered output) on the Scope and export an image of the acquisition plot (using Jingle 19.wav as input).

Set the time base to 2s/div, Channel 1 to an offset of -0.5 V and a range of 10 mV/div, Channel 2 to an offset of 0 V and a range of 1 V/div.

5. Plot the network response from the input to the output of either buffer and export an image of the bode plot. For clarity, CH1+ should be connected to WG+ at the input of the circuit, and CH2+ should be connected to the output of either buffer.

Set the sweep range to start at 10 Hz and stop at 100 kHz with 2000 samples. Configure the Wavegen, in the network tool, to an offset of 500 mV and an amplitude of 10 mV. Set the magnitude section to have a top value of 100 dB and a bottom value of 0 dB. Leave the phase settings to their default. Only plot the output, Channel 2, for amplitude and phase.

6. Plot the output spectrum of either buffer and export an image of the plot. Configure the Wavegen to output a 1 kHz, 0.5 V DC offset, 10 mV amplitude sine wave. Set the start frequency to 0 Hz and the stop frequency to 20 kHz.

Configure the magnitude so that the units are dBV, top is 20 dBV and the bottom is -80 dBV. Only plot Channel 2, the output. There should only be a large spike at 1 kHz, with minimal spikes at higher frequencies (no harmonics present).

Conclusion

Six to ten sentences (more than ten or less than six will get points off). Conclude the write up by summarizing the circuit and its performance. Weak conclusions will not be graded.